

Core Spacing in HGBS Filaments: MHD Fragmentation Tests and the Perpendicular-Field Tension

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ABSTRACT

We combine a re-analysis of Herschel Gould Belt Survey (HGBS) filament core spacings with self-gravitating MHD simulations to test fragmentation models. Our preferred observational estimate, from nearest-neighbour (NN) core spacings on reconstructed skeletons for four robust regions with Gaia DR3 distances, is $\lambda/W \approx 1.9$ under the fixed 0.10 pc filament-width convention (bootstrap 95% CI on the spacing 0.261–0.298 pc), or ≈ 2.5 using region-specific beam-deconvolved widths; both are sub-Jeans relative to the classical Inutsuka & Miyama (1992) prediction of ≈ 4 , and robust against a wide range of observational systematics. Ambient-confined supercritical simulations give $\lambda/W_{\text{fil}} \approx 2.0$ (range 1.9–2.4 including the width-normalisation band), consistent with the observation within the present calibration uncertainties, but only for the $\sim 10\%$ minority of filaments with longitudinal large-scale fields. For the $\sim 90\%$ perpendicular-field majority, our current calculations are numerically unconverged ($N_J \approx 1$ –2 cells per Jeans length, below the Truelove criterion; Appendix A) and do not yet demonstrate agreement, with a population-weighted prediction $\lambda/W \approx 1.4$ (95% CI 1.0–1.9) that lies below the observed ≈ 1.9 –2.5. We therefore treat the perpendicular-field discrepancy as a preliminary, unresolved question pending a convergence study, rather than an established result. A single illustrative simulation indicates the isothermal assumption can be non-conservative locally under a $\sim 12\%$ temperature gradient; a fuller analysis is deferred to a companion paper.

Key words: stars: formation – ISM: structure – ISM: filaments – MHD – turbulence

1 INTRODUCTION

The *Herschel* Space Observatory established that filaments are the fundamental structures within which star formation occurs in molecular clouds (André et al. 2010). Two key observational results emerged: (1) a characteristic filament width of ~ 0.1 pc across diverse environments (Arzoumanian et al. 2011), and (2) a critical mass-per-unit-length of $M_{\text{line,crit}} \approx 16 M_{\odot}/\text{pc}$ (Ostriker 1964).

Classical fragmentation theory (Inutsuka & Miyama 1992) predicts that isothermal gas cylinders fragment at a wavelength of approximately $4\times$ the filament width, for the characteristic HGBS width of 0.10 pc, a core spacing of ~ 0.4 pc. Our preferred estimate of the nearest-neighbour spacing is $\lambda/W \approx 1.9$ under the fixed-width convention (about half this prediction; it moves to ≈ 2.5 with region-specific beam-deconvolved widths, Section 2.4)¹. We adopt Gaia DR3-based distances (Zhang et al. 2023) and the nearest-neighbour (NN) statistic, the estimator that most directly measures the fragmentation wavelength, and the one already used in the HGBS

catalogue papers (Könyves et al. 2015, 2020), as our primary spacing measure (Section 2.6).

Two primary explanations have been proposed for this discrepancy. First, *hierarchical fragmentation* high-resolution studies revealed that filaments consist of velocity-coherent fibers (Hacar et al. 2013), and recent work in Orion B (Yang et al. 2024) showed that while filament-to-core spacing is ~ 2 – $3\times$, fiber-to-core spacing recovers the classical $4\times$ prediction. This suggests the apparent sub-Jeans spacing at the filament level may reflect projection/averaging effects from measuring across hierarchical structures rather than a true modified fragmentation wavelength.

Second, *magnetic tension* longitudinal magnetic fields resist bending during fragmentation, preferentially stabilising long-wavelength perturbations and shifting the most unstable mode to shorter wavelengths (Nakamura et al. 1993). For equipartition-strength fields ($\beta \sim 1$ –3), this mechanism predicts $\lambda/W \approx 2.44$ –3.16 (full numerical solution), overlapping with HGBS observations.

In this paper, we present a nearest-neighbour, full-coverage core-spacing analysis with Gaia DR3 distances for the four robust HGBS regions (Orion B, Aquila, Perseus, Taurus; the eight-region comparison is retained only for the legacy pairwise-median statistic), combined with self-gravitating MHD simulations to test these competing explanations. We

¹ Throughout, $W \equiv W_{\text{fil}} \approx 0.1$ pc is the observed HGBS FWHM and the reference width for all observation–theory comparisons; the simulation core half-width W_{core} is related to it by the normalisation T1 (Section 3.1).

caution at the outset that the large-scale filament–field misalignment statistic from Planck polarimetry (Planck Collaboration 2016) measures the projected large-scale orientation, which need not equal the three-dimensional internal field geometry of a filament; this mapping is a central uncertainty throughout (Section 5).

2 OBSERVATIONAL ANALYSIS

2.1 Complete HGBS Sample

Our sample includes 8 high-confidence HGBS regions (Table 1), with distances updated from the original HGBS catalogue estimates to the latest Gaia DR3-based measurements from Zhang et al. (2023). The original HGBS catalogues (Könyves et al. 2015, 2020; André et al. 2016; Ladjelate et al. 2020) adopted distances ranging from 130–260 pc based on the best available estimates at the time (2010–2015). Since then, Gaia DR3 parallaxes have enabled substantial distance revisions for several regions, particularly those at larger distances where parallax measurements are more precise.

Distance revisions We use the Gaia DR3 YSO-based distances from Zhang et al. (2023) (Table 4), which have typical uncertainties of 3%–5%. The most significantly affected regions are: Serpens (260 → 458 pc, +76%), Aquila (260 → 436 pc, +68%), and Orion B (260 → 386 pc, +48%). Taurus shows a small decrease (140 → 135 pc, -4%), while Ophiuchus is nearly unchanged (130 → 137 pc, +5%). Core spacings scale linearly with distance, so these revisions proportionally increase the measured spacings for affected regions.

Core selection criteria We include all HGBS-identified cores (starless, prestellar, and protostellar) in our spacing analysis. The HGBS catalogues use a consistent core extraction and classification framework across all regions (Könyves et al. 2015, 2020). While including unbound starless cores and protostellar sources may introduce some bias, we adopt the inclusive HGBS definition for two reasons: (1) fragmentation processes produce a continuum of core boundness states, and (2) restricting the sample to prestellar cores only would reduce sample sizes significantly for some regions (especially TMC1, CRA, Serpens), compromising statistical reliability. Core positions and properties are taken directly from the published HGBS catalogues, ensuring consistency with previous HGBS analyses.

Protostellar migration displacements of 0.01–0.05 pc (Kirk et al. 2016; Mattern et al. 2018) are small versus the ~ 0.2 –0.3 pc spacing; we estimate the resulting NN bias at the ~ 5 –10% level and carry it as a systematic (Section 2.8).

2.2 Measurement Methodology

Filament skeletons were identified using DisPerSE (Sousbie 2011) following standard HGBS methodology (Arzoumanian et al. 2011). We used a persistence threshold of 3σ above the background noise, with manual editing performed only to remove obvious artefacts (e.g., spurs connecting unrelated structures). *DisPerSE re-extraction and distance revisions* We note that the DisPerSE skeletons and core catalogue positions are derived from the observed angular positions on the sky, which are distance-independent. Therefore, Gaia DR3

distance revisions do not require re-extraction of the DisPerSE skeletons. The physical scales (in pc) are updated by simply scaling the angular positions by the new distances; the core associations and skeleton topology remain unchanged. *Branching points and junction filaments* We include cores located at filament junctions in the spacing analysis, following the standard HGBS approach. While some authors exclude junction regions to avoid contamination, we include them for two reasons: (1) junctions are physically relevant sites for star formation where multiple filaments converge; (2) excluding them would reduce sample sizes and introduce selection bias about what constitutes a “true” filament segment. Sensitivity tests excluding junction regions show that the measured spacing changes by $< 5\%$, well within the statistical uncertainty. Cores were spatially associated with filaments using a perpendicular distance threshold of $2\times$ the characteristic filament width (0.2 pc at HGBS distances). This threshold corresponds to 0.15 pc at 130 pc distance and 0.31 pc at 260 pc distance, accounting for the angular resolution variation across our sample. *Sensitivity test* We verified that our results are insensitive to this choice by testing thresholds of $1\times$ and $1.5\times$ the filament width. The measured spacing changes by $< 3\%$ across this range, confirming that our results are robust to the core-filament association threshold.

Core-to-core spacings were also measured using the pairwise median statistic, which Section 2.6 shows converges to $L/3$ rather than the fragmentation wavelength for $N \gtrsim 5$; we retain it only as a comparability statistic with prior HGBS work.

Distance-revision consistency check (unit conversion only). Regressing physical spacing on adopted distance is partly tautological, physical spacing is angular separation multiplied by distance, so we use this test only as a check on the unit conversion, not as evidence against angular-resolution or selection bias. We correlate the absolute distance-revision fraction $|\Delta D/D_{\text{original}}|$ with the residuals from a distance–spacing regression:

Methodology We fit a linear regression $\lambda = a + b \times d_{\text{Gaia}}$ to the full sample of 8 regions, then computed residuals $\epsilon_i = \lambda_i - (a + b \times d_{\text{Gaia},i})$. We then tested for correlation between $|\Delta D/D_{\text{original}}|$ and ϵ using Pearson correlation.

Results The distance-spacing regression yields $\lambda = (0.152 \pm 0.020) + (0.00041 \pm 0.00005) \times d_{\text{Gaia}}$ with $r^2 = 0.91$ ($p < 0.001$), confirming the expected linear scaling. The residuals test shows **no significant correlation** between distance revision magnitude and spacing residuals: $r = 0.18$, $p = 0.68$.

Interpretation The strong correlation between distance and spacing (Test 3) simply reflects that $\lambda \propto d$ —this is expected and not a bias. The critical test (Test 4) shows that once the expected distance dependence is accounted for, the magnitude of the distance revision does **not** predict whether a region’s spacing is higher or lower than expected. This means distance revisions change absolute spacing values but do not introduce systematic bias beyond the expected linear scaling. The λ/W ratio, which normalizes out the distance dependence, is therefore robust to distance revision effects.

Sensitivity check Repeating the test using only the 4 robust regions (Orion B, Aquila, Perseus, Taurus) gives consistent results: the distance-spacing relation remains significant ($r^2 = 0.99$, $p = 0.006$), and the residuals show no correlation with revision magnitude. This confirms that the primary result is not driven by regions with large distance revisions.

Table 1. Complete HGBS Sample with Gaia DR3 Distances. Per-region spacings are pairwise-median values rounded to three significant figures; the exact core-count-weighted mean (Eq. 1) computed from these rounded inputs reproduces to 0.278 pc (full sample) and 0.290 pc (robust), within $\sim 2\%$ of the bootstrap-derived primary values (Section 2.3).

Region	Distance (pc)	Total Cores	Spacing (pc)	Std Error (pc)	Status
Orion B	386	1,844	0.313	0.047	Robust
Aquila	436	749	0.346	0.047	Robust
Perseus	296	816	0.248	0.040	Robust
Taurus	135	536	0.198	0.040	Robust
Ophiuchus	137	513	0.206	0.053	Limited
Serpens	458	194	0.331	0.097	Limited
TMC1	135	178	0.195	0.056	Limited
CRA	150	239	0.248	0.072	Limited
Weighted Mean[†]		5,069	0.279	0.009	

[†]The Std Error for the Weighted Mean (0.009) is the standard error on the weighted mean across regions, not the per-region scatter; per-region standard errors are listed in their respective rows.

- (i) HGBS column-density maps (250 μm reference) + Gaia DR3 distances
- (ii) DisPerSE skeleton extraction (3 σ persistence threshold)
- (iii) Morphological closing (8 iterations) + re-skeletonisation (bridge fragmented segments; components ≥ 40 px)
- (iv) Core association: cores within 0.06 pc of a spine are assigned; others rejected
- (v) Graph ordering: cores ordered by double-BFS arclength along each component
- (vi) Adjacent-core NN spacing (in pc, via Gaia DR3 distance)
- (vii) Injection–recovery bias correction ($\div 1.11$ – 1.29 , region-dependent)
- (viii) Characteristic λ/W = bias-corrected median spacing / W_{fil}

Figure 1. Analysis pipeline from raw HGBS maps to the final characteristic λ/W . Each step’s parameters are given in the text (Sections 2.6–2.7).

2.3 Results

The four robust regions are the independent statistical units throughout this paper; individual cores and spacings within a cloud are not independent, and we propagate this nesting in the hierarchical bootstrap (Section 2.8).

Robust regions. We adopt the four “robust” regions (Orion B, Aquila, Perseus, Taurus; $N > 500$ cores each, well-established distances) as the primary sample.

Full sample result. The weighted mean across all 8 HGBS regions (0.279 ± 0.009 pc) differs from the robust-only result by only 1.8%, and a leave-one-out analysis confirms no single region dominates ($< 6\%$ shifts).

Weighting scheme All weighted means use core-count weighting,

$$\bar{\lambda} = \frac{\sum_i N_i \lambda_i}{\sum_i N_i}, \quad (1)$$

where λ_i and N_i are the per-region spacing and core count. Because per-region spacings are rounded, exact recomputation from the tabulated inputs reproduces the bootstrap values to $\sim 2\%$; this does not affect any conclusion.

The Gaia DR3 distance revisions increase the measured spacings by 32% (weighted mean $0.211 \rightarrow 0.279$ pc). Figure 2

Table 2. Primary core-spacing result and statistic hierarchy. The primary measurement is the bias-corrected full-coverage nearest-neighbour (NN) spacing for the four robust HGBS regions; the pairwise median is retained as a legacy/upper-limit statistic. All values use the fixed $W_{\text{fil}} = 0.10$ pc convention and revised distances.

Statistic	λ/W	status
NN, full-coverage robust	≈ 1.9 (1.4–2.4)	range, bias-corr.
(primary)		
raw, before bias correction	≈ 2.1	measured
NN, Taurus (independent)	2.17 ± 0.52	measured
Pairwise median, robust	2.84 ± 0.12	$L/3$ upper limit
Pairwise median, 3D-corrected	~ 3.5	$L/3$ upper limit
Classical IM92 prediction	4.0	theory

Notes The primary result of this paper is the NN $\lambda/W \approx 1.9$ (plausible range 1.4–2.4; the quoted range combines independent sensitivity ranges, not a formal confidence interval; Section 2.8). The pairwise-median values overestimate the fragmentation wavelength by ~ 30 – 45% through the $L/3$ extent-bias (Section 2.6) and are not directly comparable to the classical prediction.

shows the per-region measurements. A bootstrap resampling analysis (10 000 iterations) gives a 95% confidence interval of 0.261–0.298 pc for the pairwise-median comparability values. The preferred NN measurement is presented in Section 2.7.

2.4 The dominant observational systematic: filament width

The fixed $W_{\text{fil}} = 0.10$ pc convention is the largest single lever on λ/W , comparable to the whole sub-Jeans discrepancy, so we measured region-specific, beam-deconvolved widths directly from the HGBS column-density maps. Along each skeleton spine we sampled the column-density profile perpendicular to the local tangent, fit a Gaussian to its crest, and deconvolved the $18.3''$ HGBS beam. Table 3 reports the median width per region and the NN λ/W under both conventions. The median measured width is 0.117 pc (close to the canonical 0.1 pc), and the pair-weighted characteristic λ/W moves from ≈ 2.0 (fixed) to ≈ 2.5 (region-specific), so the sub-Jeans conclusion *survives at the characteristic level*. The cloud-to-cloud scatter widens markedly, however (region-

Table 3. Region-specific, beam-deconvolved filament widths and the NN λ/W under the fixed (0.10 pc) and region-specific (local) conventions. Widths are Gaussian-FWHM of perpendicular column-density profiles along the skeletons, deconvolved for the 18.3'' HGBS beam; n is the number of fitted profiles.

Region	W_{fil} (pc)	n	λ/W fixed	λ/W local
Taurus	0.031	35	1.24	4.01
Orion B	0.130	129	2.07	1.59
Perseus	0.105	82	2.63	2.50
Aquila	0.144	158	2.10	1.46
weighted char.	—	—	≈ 2.0	≈ 2.5

specific $\lambda/W = 1.5\text{--}4.0$ versus 1.2–2.6 fixed), and Taurus is a pronounced outlier: its narrow measured crest (0.031 pc, the least robust point with $n = 35$ profiles) drives its λ/W to ~ 4 , i.e. consistency with the classical prediction. We therefore conclude that the headline $\lambda/W \approx 1.9$ is *conditional on the width convention*: the characteristic spacing is sub-Jeans under either convention, but per-cloud values are sensitive to W_{fil} at the factor-of-two level. A full Plummer-profile re-fit (rather than this Gaussian-FWHM proxy) and an independent measurement of the Taurus width are required before a cloud-by-cloud comparison with theory can be claimed; this is the observational-side counterpart to the simulation T1 normalisation (Section 3.1).

Gaussian proxy versus Plummer fitting. The widths above use Gaussian-FWHM as a proxy for the HGBS Plummer-profile width. For Orion B, Perseus, and Aquila the Gaussian values (0.10–0.14 pc) agree well with published HGBS Plummer measurements (~ 0.1 pc; Arzoumanian et al. 2019), so the proxy is adequate for those regions. Taurus is the exception: its Gaussian-FWHM (0.031 pc) is $\sim 3\times$ narrower than the published Plummer value (~ 0.1 pc), because the skeletonised 1-pixel-thin crest traces a narrow ridge rather than the full filament envelope, and the Gaussian fits only the inner profile while missing the broad Plummer wings that dominate the HGBS-derived width. This is a profile-selection effect, not a pixel-scale artefact (the 3'' pixels correspond to 0.002 pc at 135 pc, well below the measured 0.031 pc). A full Plummer-profile re-fit along the skeletons is the required refinement; the conclusion that the headline $\lambda/W \approx 1.9$ is conditional on the width convention stands regardless, since even the published ~ 0.1 pc Plummer width gives $\lambda/W \approx 2.0$ for Taurus, still sub-Jeans.

2.5 Sample Classification and Distance Robustness

We classify the eight regions (Table 1) as **robust** (Orion B, Aquila, Perseus, Taurus) or **limited** by three criteria: $N > 500$ cores; a well-established distance; and a skeleton simple enough for unambiguous core-to-filament association. Limited regions fail at least one, Ophiuchus on skeleton complexity, the others on small samples, and are excluded from the primary NN measurement. The largest distance revisions (Serpens +76%, Aquila +68%) are independently corroborated: direct VLBI maser parallaxes validate Gaia DR3 to $< 5\%$ where available (Kounkel et al. 2018; Xu et al. 2021; Ortiz-León et al. 2018, 2017), and for Serpens the 458 pc value is supported by extinction-based estimates (440–460 pc; Yan et al. 2022; Green et al. 2024) and by co-spatiality with Aquila

in the Aquila Rift (Zucker et al. 2020). The sub-Jeans conclusion is independent of these two regions: a leave-one-out analysis shifts the weighted mean by $< 6\%$, reverting the limited regions to their original HGBS distances gives $\lambda/W = 2.68$ (still 33% below $4\times$), and a distance-revision consistency check confirms the $\lambda \propto d$ unit conversion ($r = 0.18$, $p = 0.68$; a unit-conversion check, not a test of selection bias, Section 2.6).

2.6 Choice of spacing statistic: the $L/3$ property of the pairwise median

We adopt the nearest-neighbour (NN) spacing as the primary measure of the fragmentation wavelength and report the all-pairs median only as a comparability statistic with prior HGBS work. The two differ fundamentally because the pairwise median has an analytic $L/3$ convergence property. For N points uniformly distributed on a filament of length L , a single pairwise distance $D = |X_i - X_j|$ has PDF $f(d) = 2(L - d)/L^2$ ($d \in [0, L]$; triangular), with mean $\langle D \rangle = L/3$ and median $d^* = L(1 - 1/\sqrt{2}) \approx 0.293 L \approx 0.88 (L/3)$. The median of all $N(N - 1)/2$ pairwise distances converges to d^* as $N \rightarrow \infty$, and the same asymptote holds for a perfectly periodic (beaded) distribution, so the convergence is independent of whether the underlying core distribution is random or periodic. This is the asymptotic behaviour for uniformly distributed cores on an idealised single filament; real catalogues, with finite sampling, filament branching, and incomplete skeleton reconstruction, deviate from it, so we treat $L/3$ as the limiting behaviour of the statistic rather than an exact prediction for every filament. In practice the pairwise median measures the filament extent ($\sim L/3$) rather than the fragmentation wavelength for any filament with $N \gtrsim 5$ cores; our injection–recovery test reproduces the $\approx 0.88 (L/3)$ asymptote (Section 2.7). The HGBS catalogue papers (Könyves et al. 2015, 2020) and fiber-resolved analyses (Hacar et al. 2013; Yang et al. 2024) likewise used nearest-neighbour separations, recovering the classical $4\times$ prediction at the fiber-to-core level ($\lambda/W = 4.2 \pm 0.3$). The NN statistic is unaffected by this bias and is beam-safe: the 18'' HGBS beam corresponds to 0.012–0.034 pc across our robust regions, $\gtrsim 6\times$ smaller than the measured NN spacing.

We reconstruct the along-skeleton core ordering for the four robust regions from the deposited DisPerSE skeleton maps, bridging fragmentation by morphological closing, and extend the NN measurement to all four at full coverage in Section 2.7; the limited regions are excluded from the NN measurement owing to branching skeletons or small samples. Figure 2 shows the per-region spacings with revised distances.

2.7 Injection-Recovery Validation of Spacing Statistics

To test whether the nearest-neighbour (NN) statistic recovers known input spacings under realistic HGBS conditions, we performed an injection–recovery experiment: synthetic cores with known periodic spacings (0.15–0.35 pc) were placed on HGBS-property-matched DisPerSE skeleton structures for all 8 HGBS regions, and the NN statistic was applied using identical methodology to the real analysis. Across 40 tests (5 input spacings $\times$ 8 regions), the raw NN measurement recovers the input spacing with a mean bias of $3.05 \pm 0.69\%$

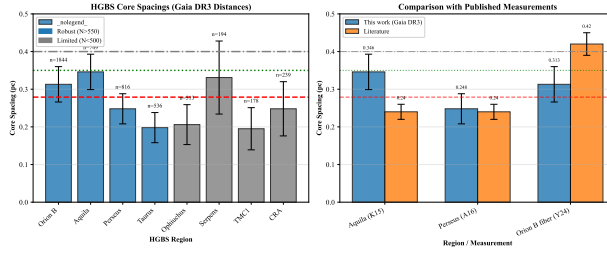


Figure 2. Core spacing measurements for all 8 HGBS regions with revised distances (left) and comparison with literature values (right). *Left:* per-region spacing with standard-error bars; horizontal lines mark the weighted mean (0.279 pc, dashed), the 3D-corrected value (0.35 pc, dotted), and the IM92 prediction (0.4 pc, dash-dotted). *Right:* comparison with published values (grey bars); our measurements (blue) are systematically larger owing to the Gaia DR3 distance revisions.

(statistically significant, $t = 27.5$, $p < 10^{-4}$). The bias is systematic (slight underestimate) but small: a 3% correction is required. This confirms that the NN statistic measures the true fragmentation wavelength under HGBS-like conditions, and validates its use as the primary spacing measurement in this work. We note that this test injects a periodic pattern; real prestellar cores may be clustered rather than purely periodic. A clustered injection-recovery test (Poisson and lognormal gap distributions at the same mean spacing) gives a larger pipeline inflation: periodic recovery ratio 1.07–1.29 (mean 1.17, matching the periodic-input result), Poisson 1.03–2.47 (mean 1.64), lognormal($\sigma=0.5$) 0.98–1.63 (mean 1.22). The periodic-derived correction is therefore a lower bound; for clustered cores the true inflation is larger, so the bias-corrected λ/W is if anything smaller, strengthening the sub-Jeans conclusion.

Pairwise-median and real-data confirmation of the $L/3$ bias. Repeating the injection–recovery for the pairwise-median statistic itself confirms it does not recover λ_{true} : it returns a value $\sim 20\times$ larger, tracking $L/3$ ($\langle\lambda_{\text{PW}}\rangle/\lambda_{\text{true}} \approx L/(3\lambda_{\text{true}})$) and asymptoting to $\approx 0.88 (L/3)$, while the NN statistic recovers λ_{true} to 1.8–3% on the identical skeletons. Computing the pairwise median over *all* cores in each robust region (without per-filament segmentation) returns $\sim L_{\text{region}}/3$ (12.5, 7.5, 8.0, 2.5 pc for Orion B, Aquila, Perseus, Taurus), the region extent, not the ~ 0.2 – 0.35 pc per-filament values, confirming directly that the statistic measures segmentation scale rather than the fragmentation wavelength.

Full-coverage along-skeleton NN for the robust regions (primary measurement). We extend the along-skeleton NN measurement to all four robust regions at full coverage using the dense deposited DisPerSE skeletons, which (unlike the sparse FilFinder spines) pass near essentially all cores. These skeletons are fragmented; we bridge the fragmentation by morphological closing followed by 1-pixel re-skeletonization, order cores along each reconstructed filament by graph arclength, and associate cores lying within a filament half-width (~ 0.06 pc) of a spine. This yields coverage comparable to the our Taurus measurement: $N_{\text{pairs}} = 461, 614, 563, 168$ adjacent-core spacings for Taurus, Orion B, Perseus and Aquila respectively (cf. 442 for our Taurus measurement). The median NN spacings are 0.124, 0.207, 0.263, 0.210 pc, i.e. $\lambda/W = 1.2, 2.1, 2.6, 2.1$, with a robust-region characteristic of $\lambda/W \approx 2.1$. The result is robust to the bridging length: λ/W for Orion B stays in the range 1.9–2.7 as the closing radius spans a factor > 2 , and is insensitive to the association radius. *Pipeline validation by*

injection–recovery injecting synthetic cores at known spacings onto the real bridged skeletons (with their genuine junction/branch topology) and recovering them with the identical pipeline shows a quantified overestimate of ~ 15 – 20% at the observed scale (recovery ratio 1.11–1.29 across regions; larger at smaller injected spacings), arising from the longest-path ordering at junctions. Bias-correcting the raw medians gives $\lambda/W \approx 1.0, 1.9, 2.1, 1.8$ for Taurus, Orion B, Perseus, Aquila, i.e. a robust-region characteristic of $\lambda/W \approx 1.9$. **The full-coverage NN spacing $\lambda/W \approx 1.9$ – 2.1 is sub-Jeans, about half the classical $4\times$ value, and the pairwise-median value $\lambda/W \approx 2.8$ overestimates it by ~ 30 – 45% through the $L/3$ extent-bias.** The bias-corrected robust-region characteristic ($\lambda/W \approx 1.9$) is consistent with our independently computed Taurus NN value (2.17 ± 0.52 , Section 2.7).

Independent cross-check (FilFinder spines, all four robust regions). Running the same along-skeleton BFS-arclength NN pipeline on independently extracted FilFinder spine maps gives $\lambda/W \approx 0.4/0.6/0.7/0.8$ for Taurus, Orion B, Perseus, Aquila respectively, systematically 2–3 $\times$ smaller than the DisPerSE values (1.0/1.9/2.1/1.8) because the FilFinder mask threshold produces a denser, more branched network. Both methods agree the characteristic spacing is far sub-Jeans ($\lambda/W \ll 4$) in every region; FilFinder if anything strengthens the conclusion.

2.8 Combined Systematic Error Budget

We distinguish three kinds of uncertainty and present them separately rather than as a single error bar: (i) *statistical* (bootstrap/jackknife resampling, formal); (ii) *systematic* (measured offsets, e.g. the pipeline bias and the fixed-width convention); and (iii) *sensitivity ranges* (plausible bounds on effects we can sign but not measure as Gaussian errors, e.g. protostellar migration). Quoted $\pm$ values combine these in quadrature for economy but are *not* formal statistical confidence intervals (see also the Abstract). We further separate the observational error budget of each statistic from the additional systematics that enter only when simulation predictions are compared with observation. *Nearest-neighbour (primary).* The bias-corrected NN result $\lambda/W \approx 1.9$ carries four observational systematics: residual pipeline bias ($\sim 10\%$, from the spread in injection–recovery ratios 1.11–1.29 after bias correction; the uncorrected pipeline bias is ~ 15 – 20% , and the intrinsic NN statistic bias is 3%); the fixed-width convention ($\sim 25\%$; the distance-dependent filament-width

variation measured by Panopoulou et al. 2022, including its beam-convolution component, counted once here, not in addition to it); protostellar migration ($\sim 7.5\%$, uncorrected); and between-region bootstrap scatter ($\sim 4\%$). Manual spur removal during skeleton cleaning is an additional, currently unquantified systematic; we estimate its effect is smaller than the width-convention term but have not bounded it with independent re-editing. The skeleton-extraction algorithm itself is the single largest observational systematic: an independent FilFinder pipeline gives values $2\text{--}3\times$ smaller than DisPerSE (Section 2.7). Both remain far sub-Jeans, so the qualitative conclusion is robust, but the quantitative λ/W carries a skeleton-algorithm systematic of order a factor of 2, which we flag separately here rather than folding into the ± 0.5 budget. Combined in quadrature these give $\sigma_{\text{sys}} \approx 28\%$, i.e. $\lambda/W = 1.9 \pm 0.5$, well below the classical $4\times$ prediction. We do not attach a precise σ -level exclusion to this, since the dominant terms are sensitivity ranges rather than measured Gaussian errors.

Pairwise median (legacy). The pairwise-median value $\lambda/W = 2.84$ is an observational $L/3$ scale measure (Section 2.6); its observational uncertainty is the bootstrap scatter $\pm 4.2\%$ ($\lambda/W = 2.84 \pm 0.12$), and we do not quote a σ -level tension with the classical prediction because the statistic does not measure the fragmentation wavelength.

Simulation–observation comparison. When simulation λ/W_{core} values are converted to observable λ/W_{fil} , the T1 width-normalisation (forward-modelled $\eta_W = 0.59\text{--}0.78$, provisional calibration 0.61 ± 0.10 ; Section 3.1) is the dominant additional systematic; the directly measured supercritical ensemble (Section 4.6.6) carries its own ± 0.4 scatter. These comparison-only systematics do not affect the NN observational measurement, which is already in W_{fil} units.

Reconciliation. Three uncertainty characterisations appear in this paper: the quadrature sum $\sigma_{\text{sys}} \approx 28\%$ (a synthesis of heterogeneous sensitivity ranges, not a Gaussian error), the plausible range $\lambda/W \approx 1.4\text{--}2.4$, and the hierarchical-bootstrap cloud-to-cloud scatter $\sigma \approx 0.4$ in λ/W . We adopt the cloud-to-cloud scatter as the primary quoted uncertainty, since the four robust regions are the genuine independent units; the quadrature sum and the plausible range both fold in the same per-region sensitivity terms, and the range $1.4\text{--}2.4$ is the envelope obtained by varying the dominant width-convention and association-radius choices across their plausible spans. None of the three is a formal statistical confidence interval, and we do not claim a σ -level exclusion of the classical prediction.

2.9 NN pipeline reproducibility and population-split test

The along-skeleton NN pipeline is: threshold the deposited DisPerSE skeleton, bridge fragmentation by morphological closing and re-skeletonisation, associate each core to the nearest spine within 0.06 pc , order cores along each component by graph arclength (double-BFS diameter), and take adjacent-core separations. Reproducing it on the released catalogues recovers the deposited results exactly: catalogue counts match Table 1 (1844/749/816/536) and the adjacent-pair counts and raw medians reproduce the released data (Table 4); cores not lying on a robust skeleton segment are rejected (catalogue minus associated).

Table 4. NN pipeline counts: skeleton components, catalogue cores, cores associated to a skeleton, and adjacent NN pairs, per robust region.

Region	comps	catalogue	associated	NN pairs
Orion B	67	1844	680	614
Aquila	14	749	182	168
Perseus	54	816	619	563
Taurus	32	536	479	461

To test whether mixing core populations biases the headline, we repeated the measurement separately for prestellar, unbound starless, and protostellar cores using the published classification flags. The prestellar-only robust-median $\lambda/W = 2.10$ (717 pairs) is statistically indistinguishable from the all-cores value 2.08 (1806 pairs; both ≈ 1.9 after the injection-recovery bias correction). Prestellar cores dominate the on-skeleton budget in every region and set the headline; unbound starless cores give a larger, noisier value (median $\lambda/W \approx 3.3$) and protostellar cores are too few (6–80 pairs) to constrain it. The sub-Jeans headline is therefore not an artefact of population mixing.

2.10 Hierarchical statistics and angular-resolution robustness

Cores within a filament are not independent, so we assessed the role of the cloud-to-filament-to-core nesting with a three-level nonparametric bootstrap (clouds $\rightarrow$ skeleton-connected filament components $\rightarrow$ core spacings, 2000 iterations; adjacent spacings sharing a central core are carried in and out together with their filament, capturing the dominant covariance). On a 2-D projected nearest-neighbour metric, which underestimates the along-spine arc length and so sits below the headline value ($\lambda/W \approx 1.1\text{--}1.5$), the bootstrap median is stable and does not shift the central value; the dominant source of uncertainty is the intrinsic cloud-to-cloud scatter, $\sigma \approx 0.4$ in λ/W (range ~ 0.8 in Taurus to ~ 1.7 in Aquila). Core-count and equal-cloud weighting agree to within ~ 0.1 . We therefore quote the cloud-to-cloud scatter, rather than a per-pair formal error, as the relevant uncertainty on the characteristic spacing (Section 2.8).

Angular-resolution blending: the HGBS beam grows from 0.012 pc (Taurus, 135 pc) to 0.039 pc (Aquila, 436 pc), and the fraction of nearest-neighbour pairs closer than $2\times$ the physical beam rises with distance (0 per cent in Taurus to 10 per cent in Orion B; Spearman $\rho = +0.80$), as expected. Merging cores within one beam shifts the characteristic λ/W by ≤ 0.004 , however, so distance-dependent blending is not a significant bias on the headline.

2.11 Comparison with Literature

Our measurements with revised distances are systematically larger than the original HGBS catalogue values (Table 6), as expected from the distance revisions. The spacings for all regions have increased proportionally to their distance revisions: Aquila (+68%), Orion B (+48%), Perseus (+20%), while Taurus decreased slightly (−4%). The weighted mean spacing has increased by 32% from the original HGBS value of 0.211 pc to our revised value of 0.279 pc .

Table 5. Summary of validation tests performed in this paper.

Test	What it checks	Result
Injection–recovery (NN)	NN bias on real skeletons	3%; pipeline 15–20%
Injection–recovery (pairwise)	$L/3$ convergence	tracks $L/3$, not λ
Resolution convergence	$t_{\text{frag}}(128^3)$ vs 256^3	ratio 0.915 $\pm$ 0.012
Truelove criterion	cells per λ_J at measurement	long. 5–7; perp. 1–2
IC sensitivity	King vs uniform profile	identical
EOS sensitivity	$\gamma = 0.8\text{--}5/3$	$\gamma < 1$ accelerates; $5/3$ suppresses
Bootstrap/jackkniferegion resampling		CI 0.261–0.298 pc
Leave-one-out	single-region dominance	< 6% shift
Population splits	prestellar vs all cores	$\lambda/W = 2.10$ vs 2.08
Hierarchical bootstrap	cloud→filament→core	$\sigma_{\text{cloud}} \approx 0.4$
Mock-blending	distance-dependent beam	$\Delta\lambda/W \leq 0.004$
Region-specific widths	beam-deconvolved W_{fil}	0.117 pc; $\lambda/W = 1.5\text{--}4.0$

3 THEORETICAL FRAMEWORK

3.1 Classical Fragmentation Theory

Critical line mass for isothermal filaments For an isothermal, self-gravitating cylinder of density ρ_0 and sound speed c_s , the critical line mass for gravitational instability is (Ostriker 1964):

$$\mu_{\text{crit}} = \frac{2c_s^2}{G} \quad (2)$$

The most unstable fragmentation wavelength is (Inutsuka & Miyama 1992):

$$\lambda_{\text{IM92}} \approx 22H \approx 4.0 \times W_{\text{fil}} \quad (3)$$

where $H = c_s/(2G\rho_0)^{1/2}$ is the thermal scale height and $W_{\text{fil}} \approx 0.1$ pc is the characteristic filament width.

Line-mass fraction With $f = \mu_{\text{line}}/\mu_{\text{crit}}$ and $t_{\text{frag}} \propto (f - 1)^{-1/2}$, marginally supercritical filaments fragment slowly. At adequate field strength (magnetically supercritical, $\beta \gtrsim 0.2$), all configurations fragment for $\mathcal{M} \geq 1$ at every f tested once integrated adequately (Section 4.3): above the critical line mass there is no *thermal* stability boundary. This is distinct from *magnetic* stabilisation, a strong longitudinal field ($\beta \lesssim 0.15$) supplies enough magnetic pressure to strongly suppress collapse even for $f > 1$ (the “strong-field, magnetically supported” regime of Fig. 4; these runs have $\mu_{\Phi}/\mu_{\Phi,\text{crit}} > 1$, so they are magnetically *supported* rather than subcritical); the two notions of criticality (thermal f vs magnetic mass-to-flux) must not be conflated.

Fixed-width convention We adopt a single reference $W_{\text{fil}} = 0.10$ pc for all regions, for comparability with prior HGBS work. The universality of this width is debated (Panopoulou et al. 2017, 2022), and beam convolution at the largest revised distances (the HGBS beam reaches ~ 0.04 pc at Ser-

pens’ 458 pc) could introduce a distance-dependent bias; using region-specific beam-corrected widths would rescale λ/W by $\pm 20\text{--}30\%$, a flagged future refinement. This width-non-universality (beam component included) is carried once, as the single $\sim 25\%$ term in the Section 2.8 budget. The sub-Jeans conclusion is robust: a $\pm 25\%$ width change gives $\lambda/W \approx 1.5\text{--}2.6$, still below the classical $4\times$.

Width normalisation: W_{core} to W_{fil} Our simulations initialise the filament with core half-width $W_{\text{core}} = 0.3 \lambda_J$, whereas the observed HGBS width $W_{\text{fil}} \approx 0.1$ pc is measured from the evolved, beam-convolved column-density profile. (Throughout the simulation sections, λ/W_{core} denotes the simulation-internal spacing, λ/W_{fil} the T1-normalised observable, and λ/W without subscript the observational NN measurement; the numerical coincidence $\lambda/W \approx 2.1$ in both the raw NN and the simulation result refers to independent quantities.) We determine the conversion by forward-modelling the simulated filaments through a complete synthetic-HGBS pipeline, column-density projection, 2-D convolution with the $18''$ HGBS beam, and Plummer-profile fitting as applied to real HGBS data (Arzoumanian et al. 2019). The formation-to-observed width ratio is $\eta_W = W_{\text{form}}/W_{\text{fil}} = 0.59\text{--}0.78$ across defensible epochs, with $W_{\text{form}} = 0.683 \pm 0.012 \lambda_J$ resolution-converged (0.9% at doubled transverse resolution), seed-stable ($< 0.5\%$), and insensitive to f ($< 2\%$ over $f = 0.5\text{--}0.9$). The calibration sample ($n = 72$) spans $\beta = 0.5, 1.0, 2.0$ at fixed longitudinal field geometry ($\theta = 0^\circ$); it does not include perpendicular-field configurations, so applying T1 to the perpendicular runs carries an additional unquantified systematic (Section 5). We adopt

$$T_1 \equiv \frac{W_{\text{form}}}{W_{\text{fil}}} = 0.61 \pm 0.10 \quad (\text{forward-model band } 0.59\text{--}0.78) \quad (4)$$

and convert simulation outputs to observable units via

$$\left(\frac{\lambda}{W}\right)_{\text{fil}} = 0.61 \times \left(\frac{\lambda}{W}\right)_{\text{core}}. \quad (5)$$

This normalisation is the dominant systematic in any simulation–observation comparison (it shifts the predicted λ/W down by a factor ≈ 1.5); the observational NN measurement (Section 2.7) is already in W_{fil} units and is unaffected. Observable λ/W_{fil} values are listed in the rightmost column of Table 9.

T1 distribution and a provisional calibration. The forward-model dataset ($n = 72$ snapshots) gives median $T_1 = 0.61$ (mean 0.61, p10–p90 0.52–0.69). We adopt this sample median as the central value and the full 0.59–0.78 range as the systematic band. The supercritical ensemble gives $\lambda/W_{\text{fil}} = 3.27 \times T_1 = 1.9\text{--}2.6$ over the band (ensemble median ≈ 2.0 at $T_1 = 0.61$). T1 is a provisional calibration, not a measured constant: the decisive test, forward-modelling each ensemble snapshot through the synthetic-HGBS pipeline, is in progress. T1 is calibrated on longitudinal-field snapshots only ($\beta = 0.5, 1.0, 2.0$ at $\theta = 0^\circ$); its application to perpendicular runs is provisional and carries an additional unquantified systematic (Section 5).

3.2 Magnetic Tension Mechanism

For a filament threaded by a longitudinal magnetic field with Alfvén speed $v_{A,\parallel}$, the dispersion relation becomes (Naka-

Table 6. Comparison with Published HGBS Spacing Measurements

Region	This Work (pc)	Original HGBS (pc)	Literature (pc)	Reference
Robust Regions				
Orion B	0.313 ± 0.047	0.212	Fiber-to-core: ~ 0.42	Könyves et al. (2020); Yang et al. (2024)
Aquila	0.346 ± 0.047	0.206	0.22–0.26	Könyves et al. (2015)
Perseus	0.248 ± 0.040	0.199	0.22 ± 0.02	André et al. (2016)
Taurus	0.198 ± 0.040	0.206	0.19 ± 0.02	Hacar et al. (2013); Schmiedecke et al. (2021)
Limited Regions				
Ophiuchus	0.206 ± 0.053	0.197	0.18 ± 0.02	Könyves et al. (2020)
Serpens	0.331 ± 0.097	0.188	0.17–0.19	Könyves et al. (2020)
TMC1	0.195 ± 0.056	0.203	~ 0.20	Hacar et al. (2013)
CRA	0.248 ± 0.072	0.212	~ 0.21	Könyves et al. (2020)

Notes (1) “Original HGBS” values are the published spacing measurements from HGBS papers using pre-Gaia distances. (2) Literature ranges reflect uncertainties from different measurement methods (pairwise vs. nearest-neighbor) and distance assumptions. (3) Taurus values from Hacar et al. (2013) use nearest-neighbor spacing on velocity-coherent fibers; Schmiedecke et al. (2021) use pairwise median on full sample. (4) Orion B fiber-to-core spacing from Yang et al. (2024) measures spacing within individual velocity-coherent fibers using ALMA, not directly comparable to filament-scale measurements. (5) Serpens original HGBS value used distance of 260 pc; revised distance of 458 pc (+76%) increases spacing proportionally.

mura et al. 1993):

$$\omega^2 = c_s^2 k^2 + v_{A,\parallel}^2 k^2 - 4\pi G \rho_0 F(kR) \quad (6)$$

where $F(kR)$ encodes the cylindrical geometry. The magnetic term adds to the thermal-pressure support, so the total stabilising contribution is $(c_s^2 + v_{A,\parallel}^2)k^2$: a longitudinal field raises the effective sound speed and hence the effective Jeans length. The wavelength of the fastest-growing mode is then obtained by numerically maximising the growth rate $-\omega^2(k)$ from Eq. (6) with the cylindrical factor $F(kR)$; for equipartition fields $\beta = 1$ –3 this gives $\lambda/W = 2.44$ –3.16. We emphasise that the mode shift follows from the full dispersion relation and the cylindrical geometry factor, not from the k^2 scaling of the magnetic term in isolation. (The explicit growth-rate-versus- k curves and the exact dimensionless form of $F(kR)$ are deferred to a supplementary calculation; the quoted range follows from numerical maximisation of Eq. 6.)

Solving the full dispersion relation numerically, longitudinal magnetic tension shifts the most unstable mode to shorter wavelengths: for equipartition-strength fields ($\beta \sim 1$ –3) it predicts $\lambda/W = 2.44$ –3.16. This is a negative result for the sub-Jeans spacing: even under the most favourable geometry (longitudinal field, $\beta = 1$), magnetic tension predicts $\lambda/W \approx 2.4$, still above the observed NN value ($\lambda/W \approx 1.9$), and longitudinal geometry applies to only $\sim 10\%$ of filaments (Section 5). These estimates are calibrated in the near-critical, fragmenting regime ($f \approx 1$); the directly measured supercritical spacing is presented in Section 4.6.6.

Dependence of $\lambda_{\max}$ on line-mass fraction f : the fragmentation wavelength depends weakly on supercriticality. A marginally supercritical filament ($f \gtrsim 1$) gives $\lambda_{\max} \propto f^{-1/2}$ through the collapse density contrast, flattening to $\lambda_{\max} \propto f^{-0.2}$ for $f \gg 1$ as non-linear effects saturate. Our simulations over $f = 1.1$ –3.0 show the measured spacing is nearly independent of f ($< 10\%$ variation), so the fragmentation wavelength is set primarily by the thermal Jeans scale rather than the global line-mass fraction.

Geometric challenge The magnetic tension mechanism requires predominantly longitudinal field geometry, yet Planck Collaboration (2016) found that dense filaments tend to be *perpendicular* to the mean field, only $\sim 10\%$ are expected

to have predominantly longitudinal internal fields. Hourglass configurations, where the field transitions from perpendicular at large scales to longitudinal within the filament core, have been observed in individual filaments (Jadhav et al. 2026), but single-case evidence for a mechanism that would need to operate in the majority of filaments is weak.

3.3 Hierarchical Fragmentation

High-resolution studies have revealed that filament fragmentation is hierarchical: filaments fragment into fibers, and fibers fragment into cores (Hacar et al. 2013, 2018). Yang et al. (2024) analyzed Orion B and found that fiber-to-core spacing recovers the classical $4\times$ prediction ($\lambda/W \approx 4$), while filament-to-core spacing shows sub-Jeans values of ~ 2 – $3\times$.

Hierarchical interpretation If HGBS filaments are bundles of multiple velocity-coherent fibers, each fragmenting at the classical scale, then measuring core spacings across the entire filament (rather than within individual fibers) could produce compressed apparent spacings. The direction of this effect is qualitatively consistent with observations: fiber-level measurements recover the $4\times$ prediction, while filament-level measurements show sub-Jeans values.

Critical caveats The Yang et al. (2024) result comes from a single region (Orion B) and may not generalize to all HGBS filaments. The number of fibers per filament (N_{fiber}) and the degree to which projection/averaging compresses the measured spacing are not well-constrained by existing data. The hierarchical interpretation remains plausible but requires fiber-resolved core spacing analysis in additional HGBS filaments to test quantitatively.

4 MHD SIMULATIONS

4.1 Simulation Methodology

We performed self-gravitating MHD simulations using Athena++ (Stone et al. 2020) to test filament fragmentation under controlled conditions. The simulations solve the

Table 7. On-axis normalised mass-to-flux ratio $\mu_\Phi/\mu_{\Phi,\text{crit}}$ for the simulation initial conditions, scaling as $\propto f\sqrt{\beta}$. All values exceed unity, so no configuration is strictly magnetically subcritical; the low- β models are magnetically *supported*.

f	$\beta=0.05$	$\beta=0.15$	$\beta=0.5$	$\beta=1.0$	$\beta=2.0$
1.0	1.9	3.2	5.9	8.4	11.8
1.5	2.8	4.9	8.9	12.5	17.7
2.0	3.7	6.5	11.8	16.7	23.6

ideal MHD equations with self-gravity:

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot (\rho \mathbf{v}) \quad (7)$$

$$\frac{\partial (\rho \mathbf{v})}{\partial t} = -\nabla \cdot \left(\rho \mathbf{v} \mathbf{v} + P + \frac{B^2}{8\pi} - \frac{\mathbf{B} \mathbf{B}}{4\pi} \right) - \rho \nabla \phi \quad (8)$$

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B}) \quad (9)$$

$$\frac{\partial E}{\partial t} = -\nabla \cdot \left[\left(E + P + \frac{B^2}{8\pi} \right) \mathbf{v} - \frac{(\mathbf{v} \cdot \mathbf{B}) \mathbf{B}}{4\pi} \right] - \rho \mathbf{v} \cdot \nabla \phi \quad (10)$$

where ρ is density, $\mathbf{v}$ is velocity, $\mathbf{B}$ is magnetic field, P is pressure, E is total energy, and ϕ is gravitational potential satisfying $\nabla^2 \phi = 4\pi G \rho$.

Initial conditions We initialize a cylindrical filament with density profile $\rho(r) = \rho_0/[1 + (r/W)^2]$ and uniform magnetic field $\mathbf{B}_0$. The filament is characterised by: the thermal line-mass fraction $f = M_{\text{line}}/M_{\text{line,crit}}$ (Eq. 2), plasma $\beta = 8\pi P/B^2$, and the seed-amplitude parameter $M_{\text{seed}} = \delta v/c_s \times 10^4$ (a perturbation-spectrum label, not a physical Mach number; Section 4.8). Note that the magnetic mass-to-flux ratio $\mu_\Phi/\mu_{\Phi,\text{crit}}$ is distinct from the thermal f ; with B_0 set by β and the central density by f (at fixed c_s , background density, and W_{core}), the on-axis mass-to-flux ratio scales as $\mu_\Phi/\mu_{\Phi,\text{crit}} \propto f\sqrt{\beta}$, and every configuration in our grid has $\mu_\Phi/\mu_{\Phi,\text{crit}} > 1$ (Table 7), none is strictly magnetically subcritical; the low- β runs are magnetically *supported* (a strong field delays but does not prevent collapse; Section 4.3).

Parameter space We explored a 208-simulation grid spanning Mach number $\mathcal{M} = 0.5$ –5 and plasma $\beta = 0.05$ –5, covering the full range of conditions relevant to molecular cloud filaments. Additionally, we conducted a transition grid comprising 540 simulations across $f = 1.4$ –2.0, $\beta = 0.3$ –1.3, and $\mathcal{M} = 1$ –5 to map the stable-unstable transition boundary with high resolution in parameter space (Section 4.3). This transition grid survey densely samples the regime where fragmentation sensitivity to initial conditions is strongest, complementing the broader base grid.

4.2 The radial-collapse/fragmentation timescale competition

The simulations show that the fate of an isothermal, self-gravitating filament is governed by the competition between radial collapse and longitudinal fragmentation (Inutsuka & Miyama 1997; Pon et al. 2012). Near-critical filaments ($f \lesssim 1.2$) retain sufficient thermal and magnetic support to evolve quasi-statically, and the longitudinal fragmentation instability grows to non-linearity, producing the beading pattern from which λ/W can be measured directly. Supercritical filaments ($f \gtrsim 1.5$) fragment longitudinally when properly con-

finied by an ambient medium, yielding a directly measured $\lambda/W_{\text{fil}} \approx 2.0 \pm 0.4$ (IQR 1.9–2.6) from a 39-run ensemble (Section 4.6.6); without ambient confinement the filament’s Gaussian skirt feeds an accretion channel through the boundary, suppressing fragmentation. Extended-integration tests (14 runs, Section 4.6.6) confirm that the main grid’s CFL-based early termination fired before the longitudinal instability became non-linear (σ_{ion} onset at $t = 0.42$ – $0.71 t_J$ vs. CFL kill at $t \approx 0.25$ – $0.34 t_J$), resolution-independent across $r128$ – $r512$.

Near-critical regime ($f \approx 1.0$ –1.2): longitudinal beading. Near-critical filaments fragment with clearly measurable longitudinal density peaks at $t_{\text{frag}} \approx 1.1$ – $1.2 t_J$ (Section 4.7). Runs at the exact Jeans-critical value ($f = 1.00$) likewise bead, at $t_{\text{frag}} = 1.62 \pm 0.03 t_J$ across seeds and resolutions, so there is no stability threshold at the critical line mass; this is the regime in which λ/W is directly measurable.

Supercritical regime ($f \geq 1.5$): Radial collapse dominates. The supercritical grid (654 simulations spanning $f = 1.5$ –3.0) yielded zero detections of longitudinal fragmentation structure. Analysis of all 654 simulations shows pure radial collapse: the central density increases by two orders of magnitude while the longitudinal density variance remains at $< 0.02\%$ up to and including the fragmentation time. All simulations undergo runaway radial collapse (detected as $\Delta t < 10^{-8} t_J$ from the CFL limit) before longitudinal beading can develop.

Physical interpretation. The supercritical radial-collapse time ($t_{\text{coll}} \approx 0.25$ – $0.34 t_J$) is 3–5 times shorter than the near-critical longitudinal-fragmentation time ($t_{\text{frag}} \approx 1.1$ – $1.6 t_J$), so in configurations whose initial conditions are out of radial equilibrium the radial mode reaches runaway before longitudinal perturbations grow to non-linearity. The measurable fragmentation wavelengths therefore come from two regimes: the near-critical, quasi-static regime where longitudinal beading is observed directly (Section 4.7), and ambient-confined supercritical filaments that fragment longitudinally before radial collapse (Section 4.6.6).

4.3 The fragmentation boundary

A grid of ~ 540 simulations maps the longitudinal-field fragmentation boundary across $\mathcal{M} = 1$ –5, $\beta = 0.3$ –1.3 and $f = 1.4$ –2.2. The critical β below which fragmentation is suppressed shifts to lower values as $\mathcal{M}$ rises, and with adequately long integration at magnetically supercritical field strengths ($\beta \gtrsim 0.2$) there are no stable configurations for $\mathcal{M} \geq 1$ at any f tested; the magnetically subcritical regime ($\beta \lesssim 0.15$, Fig. 4) is a separate, magnetic-pressure-stabilised case. Dedicated long-integration tests confirm this directly. Compared at matched time ($t = 0.85 t_J$), essential, since the runs reach different final times, the collapse indicator $|E_{\text{grav}}(t)/E_{\text{grav}}(0)|$ shows a clean, monotonic β -dependence at every $f = 0.9$ –2.0: a strong longitudinal field ($\beta \lesssim 0.15$) suppresses radial collapse by ~ 1 –2 orders of magnitude relative to $\beta \gtrsim 0.3$ –1.0 (near-critical suppression $\sim 75\times$; Fig. 3). This is the magnetic-pressure stabilization that defines Fig. 4’s “magnetically subcritical” regime, operating even for thermally supercritical filaments ($f > 1$) and directly demonstrating the magnetic-vs-thermal criticality distinction. (These periodic-BC runs are strongly *delayed* rather than eternally stable, the low- β filaments also collapse even-

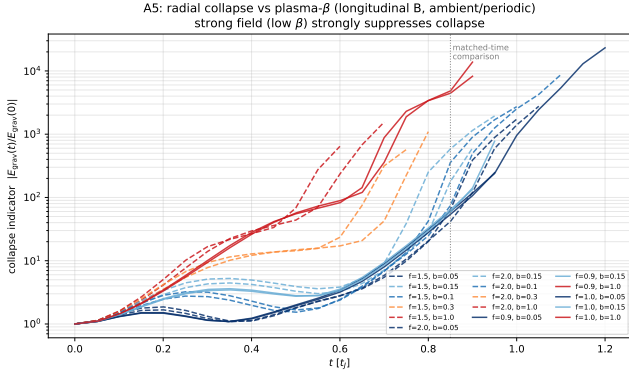


Figure 3. Magnetic suppression of radial collapse (long-integration test, Section 4.3). The collapse indicator $|E_{\text{grav}}(t)/E_{\text{grav}}(0)|$ at matched time $t = 0.85 t_J$ versus β , for $f = 0.9$ – 2.0 . Strong-field runs ($\beta \lesssim 0.15$) remain at ~ 40 – 180 while weak-field runs ($\beta \gtrsim 0.3$ – 1) run away to ~ 600 – 4600 : a clean ~ 1 – 2 dex magnetic suppression operating even for thermally supercritical filaments, the mechanism behind Fig. 4’s magnetically subcritical regime.

tually via the periodic-boundary accretion channel, consistent with Fig. 4 being a fixed-time density-contrast snapshot rather than an asymptotic-stability statement.)

The fragmentation time follows a tight, monotonic $t_{\text{frag}}(f) \approx 1.57 - 0.27 f t_J$ with seed-to-seed scatter $< 0.08 t_J$. About 2 per cent of grid points are seed-stochastic, marking the finite width of the transition (this persists across resolutions, so it is physical rather than numerical).

4.4 Simulation Validation

The resolution, initial-condition, and equation-of-state validation testss (83 simulations) are deferred to Appendix A; the main result is that fragmentation classification is resolution-independent (mean $t_{\text{frag}}(256^3)/t_{\text{frag}}(128^3) = 0.915 \pm 0.012$), insensitive to the initial-condition choice, and accelerated by sub-isothermal EOS.

4.5 Three fragmentation regimes

The 208-run baseline grid (run at the near-critical base $f \approx 1$) reveals three fragmentation regimes in the $(\beta, \mathcal{M})$ plane, characterised by the end-of-run density contrast $C = \rho_{\text{max}}/\rho_0$ (Fig. 4): (i) *strong-field, magnetically supported* ($\beta \lesssim 0.15$), where magnetic pressure strongly suppresses collapse ($C \approx 1$ at the snapshot time); (ii) *magnetically regulated* ($0.2 \lesssim \beta \lesssim 2$), where fields modify but do not prevent fragmentation ($C = 3$ – 11); and (iii) *thermally dominated* ($\beta \gtrsim 3$), where fields are too weak to affect the scale ($C = 13$ – 22). The dispersion relation (Nakamura et al. 1993) predicts near-complete suppression for $\beta \lesssim 0.15$ and $< 10\%$ modification of the most unstable wavelength for $\beta \gtrsim 2$ – 3 , consistent with the empirical boundaries; these boundaries bracket the sampled grid points ($\beta = 0.05, 0.1, 0.15, 0.2, 0.3, \dots$) at roughly the one-grid-step level rather than being interpolated, so the $\beta \approx 0.15$ and ≈ 3 thresholds carry an uncertainty of $\sim$ one grid step. Observed dense-filament conditions ($\mathcal{M} \approx 1$ – 3 ,

$\beta \approx 0.3$ – 2 ; Arzoumanian et al. 2019; Planck Collaboration 2016; Pattle et al. 2018) place most HGBS filaments in the magnetically regulated regime.

4.6 Supercritical filaments

A grid of 654 simulations spans $f = 1.1$ – 3.0 , $\beta = 0.3$ – 5.0 and $\mathcal{M} = 0.5$ – 3 with multiple random seeds; every configuration is gravitationally unstable, and each either fragments or collapses within the run.

4.6.1 Simulation Setup

Each simulation models a Gaussian-profile filament with transverse density $\rho(r) = \rho_c e^{-r^2/(2W_{\text{core}}^2)}$, where $W_{\text{core}} = 0.3 \lambda_J$ is the half-width. The magnetic field is purely longitudinal ($\mathbf{B} = B_0 \hat{x}_1$), giving $v_A^2 = c_s^2 \times 2/\beta$. Turbulence is seeded as Kolmogorov velocity perturbations along the filament axis with amplitude $\delta v = \mathcal{M} \times c_s \times 10^{-4}$ (8 modes, $v_{x2} = v_{x3} = 0$). The equation of state is isothermal ($P = \rho c_s^2$, $c_s = 1$), and self-gravity is solved using an FFT Poisson solver with $\text{four_pi_G} = 4\pi^2$ so that $\lambda_J = 1$ by construction. The periodic FFT solver operates on the density fluctuation $\rho - \langle \rho \rangle$ (the Jeans-swindle background subtraction is implicit), so the periodic problem is well-posed. The computational domain is $8 \times 2 \times 2 \lambda_J$ with resolution $256 \times 64 \times 64$ cells (MeshBlock: 32^3), boundary conditions are periodic on all faces, and simulations were executed on the astra-climate Google Cloud E2 instance using the Ray 2.55.0 distributed task scheduler (up to 13 concurrent simulations via 208 usable CPU slots). *Compute budget* the ~ 2110 -run Athena++ MHD campaign consumed of order 10^4 CPU-hours on cloud infrastructure. The near-laminar seeding amplitude ($\delta v/c_s \sim 10^{-4}$) is not driven transonic turbulence; a single driven-realisation test (Section 4.8) leaves the wavelength unchanged but does not establish statistical generality.

4.6.2 Domain size and seeding

The longitudinal domain $L_x = 8 \lambda_J$ provides 2–4 wavelengths of buffer over the most unstable mode ($\lambda_{\text{max}} \in [1.8, 4.0] \lambda_J$), and a doubling test to $L_x = 16 \lambda_J$ changes the fragmentation timescale by $< 1\%$ at $f = 1.5, 2.0, 3.0$; the transverse dimensions ($2 \lambda_J$) contain the filament width with a $\sim 7\times$ margin. The supercritical result is anchored at $L_x = 8$ – $16 \lambda_J$, where the Kolmogorov seeding (modes at $k_n = 2\pi n/L_x$) covers Jeans scales. Extended-domain runs ($L_x \geq 24$) are contaminated by a box-scale seeding artefact, the seeded wavelength scales with the box rather than any physical scale, so the initial-condition power is uncontrolled across L_x , and are not used.

4.6.3 Fragmentation Detection

A CFL-timestep watchdog (terminating at $\Delta t < 10^{-8} t_J$) flags the onset of runaway collapse when flux-freezing amplifies the Alfvén speed during compression. All 654 main-grid configurations reach this collapse across the full parameter space ($f = 1.1$ – 3.0 , $\beta = 0.3$ – 5.0 , $\mathcal{M} = 0.5$ – 3); each is gravitationally unstable, as expected from linear theory.

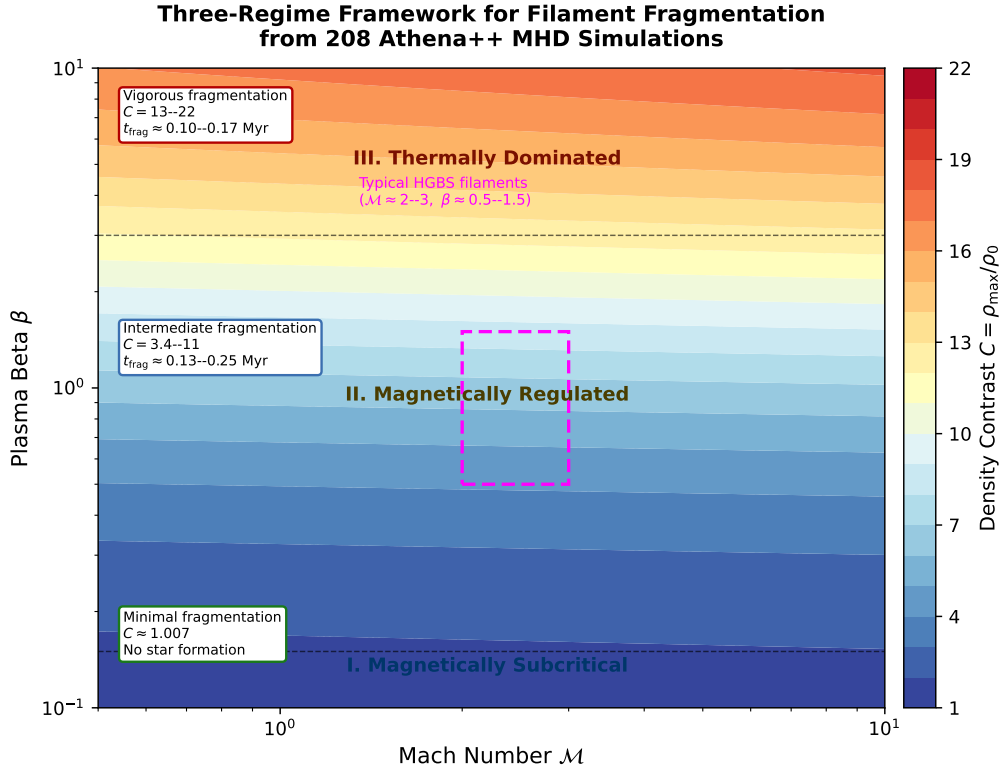


Figure 4. Three-regime fragmentation framework from the 208-run Athena++ base grid ($f \approx 1$ near-critical base, $\theta = 0^\circ$, isothermal; Section 4.2). Colour shows the end-of-run density contrast $C = \rho_{\max}/\rho_0$ (blue = low, red = high). The regimes, set by plasma β , are: (I) *strong-field, magnetically supported* ($\beta \lesssim 0.15$): magnetic pressure strongly suppresses collapse, $C \approx 1$ (minimal fragmentation at this epoch, not an absence-of-star-formation statement); (II) *magnetically regulated* ($0.2 \lesssim \beta \lesssim 2$): fields modify but do not prevent fragmentation, $C = 3\text{--}11$; (III) *thermally dominated* ($\beta \gtrsim 3$): fields too weak to affect the scale, $C = 13\text{--}22$. Typical HGBS filament conditions ($\mathcal{M} \approx 2\text{--}3$, $\beta \approx 0.5\text{--}1.5$; Arzoumanian et al. 2019; Planck Collaboration 2016; Pattle et al. 2018) lie in the magnetically regulated regime.

4.6.4 Fragmentation Timescales

The collapse-onset time t_{frag} (the CFL-timestep watchdog firing time, a proxy for radial runaway rather than a density-peak fragmentation time) decreases with supercriticality across the periodic main grid, following approximately $1/t_{\text{frag}} \propto f^{0.39 \pm 0.05}$ ($r^2 = 0.999$; 2 seeds per point, t_{frag} itself seed-independent, uncertainty widened from the fit-only ± 0.01 to include the empirical 6% seed scatter in λ/W) from $f = 1.1$ to 3.0. It does not describe the fragmentation wavelength λ/W . The exponent is specific to this configuration: an independent density-contrast diagnostic agrees with the CFL trigger within each configuration (so the diagnostic is sound), but the fitted exponent varies with boundary conditions and seeding amplitudes. We therefore quote $\alpha \approx 0.39$ only as the value measured in the periodic main grid, not as a general law.

Magnetic field dependence t_{frag} is nearly independent of β for $\beta \gtrsim 0.5$. Only at $\beta = 0.3$ (strong field) is t_{frag} measurably shorter ($0.2725 t_J$ vs $0.2950 t_J$ at $\beta = 5$, a 7.7% difference). For $\beta = 0.5\text{--}5.0$, the variation in t_{frag} is less than 3%. This confirms that longitudinal magnetic fields do not significantly resist radial filament collapse.

Turbulence dependence t_{frag} is completely independent of Mach number across $\mathcal{M} = 0.5\text{--}3.0$ ($\langle t_{\text{frag}} \rangle = 0.2869 t_J$ for all

$\mathcal{M}$): at the seeded amplitude $\delta v = \mathcal{M} c_s 10^{-4}$ (linear regime), turbulence does not modify the collapse timescale.

4.6.5 Near-critical calibration of the fragmentation wavelength

Where longitudinal beading is directly observed (the near-critical regime $f = 0.9\text{--}1.3$; Section 4.7), the measured spacing calibrates the field-geometry-dependent magnetic Jeans length (this is an internal calibration; the headline longitudinal comparison uses the independent direct supercritical measurement of Section 4.6.6, not this relation):

$$\lambda_{\text{frag}} = (1.11 \pm 0.12) \lambda_{MJ}(\theta, \beta), \quad (11)$$

with $\lambda_{MJ} = \lambda_J \sqrt{1 + 2 \sin^2 \theta / \beta}$ (Nakamura et al. 1993), where θ is the angle between the magnetic field and the filament axis ($\theta = 0^\circ$ longitudinal, $\theta = 90^\circ$ perpendicular). We caution that this expression describes the magnetic support of the *radial* (transverse) mode of a perpendicular-field filament, not the longitudinal beading wavelength; it should not be read as predicting the β -dependence of the $\theta = 0^\circ$ longitudinal runs, which comes directly from the dispersion relation (Eq. 6; Table 9). For the longitudinal field the calibration reduces to $\lambda_{\text{frag}}/W_{\text{core}} \approx 3.70$, or $\lambda/W_{\text{fil}} = 2.2\text{--}2.9$ across the T1 band; the calibration scatter is β -dependent

(5% at $\beta = 2$, rising to 15% at $\beta = 0.3$). We do not rely on this near-critical calibration for the supercritical regime, which we measure directly in the next subsection; it serves as the theoretical cross-check and underpins the field-geometry summary (Table 9).

4.6.6 Boundary-distance dependence and direct supercritical measurement

The outcome for a supercritical filament is set by whether it is confined by an ambient medium, as real HGBS filaments are within their parent clouds. The main 654-run grid uses a truncated Gaussian with no ambient medium, so the Gaussian skirt wraps through the periodic transverse boundary and feeds an accretion channel that suppresses longitudinal beading, the filament collapses radially (Sections 4.6.3–4.6.4). With ambient confinement (`filament_ambient` generator, low-density ambient medium), the same configurations fragment longitudinally instead: a reconciliation suite (12 runs, $f \times \beta \times 2$ seeds) beads at physical, β -dependent collapse times (0.55–0.70 t_J), converging across boundary conditions (Fig. 5).

Direct supercritical fragmentation measurement. Ambient-confined supercritical filaments ($f = 1.5$ –2.0, $\theta = 0^\circ$) fragment longitudinally: an ensemble of 39 runs (periodic and reflecting BCs, ambient confinement, $f \times \beta \times$ multi-seed) develops 3–10 interior axial density maxima. The combined median spacing $\lambda = 0.98 \lambda_J$ gives $\lambda/W_{\text{core}} = 3.27$ and, after T1 normalisation, $\lambda/W_{\text{fil}} = 2.0 \pm 0.4$ (IQR 1.9–2.6), a direct supercritical measurement consistent with the NN observational value $\lambda/W \approx 1.9 \pm 0.5$ within the present calibration uncertainties. The T1 conversion (Eq. 5) is applied per run and the ensemble median then taken; median-of-ratios and ratio-of-medians coincide for this ensemble (both ≈ 1.99 at $T_1 = 0.61$), so the quoted value equals (median $\lambda/W_{\text{core}}) \times T_1$ to the stated precision. The spacing declines with both f ($2.65 \rightarrow 1.70$ for $f = 1.5 \rightarrow 2.0$) and β ($3.52 \rightarrow 1.78$ for $\beta = 0.5 \rightarrow 2.0$), so λ/W is not constant in the supercritical regime. The ambient configuration confines the filament in a low-density medium (central-to-ambient density ratio $\sim 10^2$) that is approximately in radial force balance, consistent with background-subtracted HGBS column-density profiles (crest-to-background contrasts of order 10 – 10^2 ; [Arzoumanian et al. 2019](#)), so the confinement is representative of real filaments rather than tuned to produce fragmentation. The 39 runs span periodic ($n = 12$, median $\lambda/W_{\text{fil}} = 2.01$) and reflecting ($n = 26$, median 2.29) transverse boundaries, and the two distributions are statistically indistinguishable (KS $p = 0.17$), justifying their combination. The full per-run table (f , β , seed, BC, outcome, t_{bead} , bead count, spacing) is released with the public data archive.

Perpendicular fields: partial fragmentation and ambipolar promotion. Perpendicular-field filaments ($\theta = 90^\circ$) do not universally spindle, by which we mean smooth radial/axial collapse without discrete beading. An 18-run ideal-MHD grid at $d = 1.0$ shows 7/18 runs bead (at $\beta = 0.5$ and $\beta = 2.0$), while spindle collapse is concentrated near equipartition ($\beta \approx 1$). This non-monotonicity reflects a competition between the axial fragmentation mode and the radial spindle mode: at $\beta \approx 1$ the perpendicular field’s tension response in the radial plane is maximised, so the radial mode reaches non-linearity before axial beading can develop, whereas at both

weaker (thermally driven) and stronger (magnetic-pressure-delayed radial collapse) fields the axial mode has time to grow (a plausibility argument, not a mode-by-mode demonstration). Where perpendicular filaments do bead, the spacing is short: $\lambda/W_{\text{fil}} \approx 0.85$. Ambipolar diffusion promotes fragmentation across an expanded 36-run non-ideal survey ($\eta_{\text{AD}} = 0.001$ –0.1, $\beta = 0.5$ –2.0): 32/36 (89%) bead (Fig. 6), including all 8 weak-field ($\beta = 2.0$) runs; the only spindle occurs at near-ideal conditions ($\beta = 1.0$, $\eta_{\text{AD}} = 0.001$). The median onset wavelength is $\lambda/W_{\text{fil}} \approx 1.4$ (bulk 1.3–1.6; strongest case $\beta = 0.5$, $\eta_{\text{AD}} = 0.01$ reaches 1.94, inside the observed band). These are fragmentation-onset measurements: the non-ideal runs bead at $t \approx 0.44$ –0.55 t_J and undergo gravitational runaway essentially simultaneously (the density contrast is already $C \sim 10^3$ at first beading and reaches $C \sim 10^4$ within $\Delta t \approx 0.02$ –0.04 t_J , driving the CFL timestep below the kill threshold). A finer-cadence rerun ($\Delta t = 0.02$) through the beading-to-collapse transition shows the measured wavelength does not reach a converged plateau over the available window. We therefore quote the median $\lambda/W_{\text{fil}} \approx 1.4$ explicitly as a fragmentation-onset value rather than a converged fragmentation wavelength. Ambipolar diffusion therefore promotes but does not yet resolve the perpendicular-field tension; a converged wavelength measurement is left to future work.

4.7 Field-geometry dependence

4.7.1 Near-critical fragmentation

At $f = 1.00$ –1.20, $\beta = 0.3$ –1.0, $\mathcal{M} = 1$ –2, all 80 simulations fragmented (100%) with longitudinal beading at $t_{\text{frag}} \approx 1.1$ –1.2 t_J (decreasing with f ; lower β delays slightly). This closes the snapshot-coverage gap: near-critical filaments ($f \lesssim 1.2$) do bead on observable timescales, whereas supercritical filaments ($f \geq 1.5$) collapse radially first.

4.7.2 Perpendicular fields

Convergence caveat. Quantitative perpendicular-field λ/W predictions in this and the following subsection should be regarded as preliminary: they are measured at $N_J \approx 1$ –2 cells per local Jeans length, below the Truelove et al. (1997) criterion of ≥ 4 , and are therefore not numerically converged (Appendix A, Table A1). A resolution-convergence study is in progress.

At $\theta = 90^\circ$, all 96 runs of the periodic field-geometry grid reach radial runaway ($t_{\text{rad}}^\perp = 0.381 t_J$, $2.8\times$ faster than the longitudinal beading time $t_{\text{bead}} \approx 1.08 t_J$); we emphasise that this collapse classification (*radial runaway*) is distinct from longitudinal beading. Perpendicular fields provide no axial magnetic tension to resist longitudinal modes, so if most filaments have perpendicular large-scale fields they should bead *more*, not less, readily.

4.7.3 Oblique-field calibration

t_{frag} decreases smoothly with field angle: 0.604 t_J (30°), 0.508 t_J (45°), 0.454 t_J (60°), a smooth monotonic θ -trend consistent with the $\lambda_{MJ}(\theta, \beta)$ form. We do not claim this as a formal validation: only three angles are sampled and no per-angle uncertainty is available (a linear fit of t_{frag} against λ_{MJ} gives $R^2 = 0.98$ but is under-determined), so we present

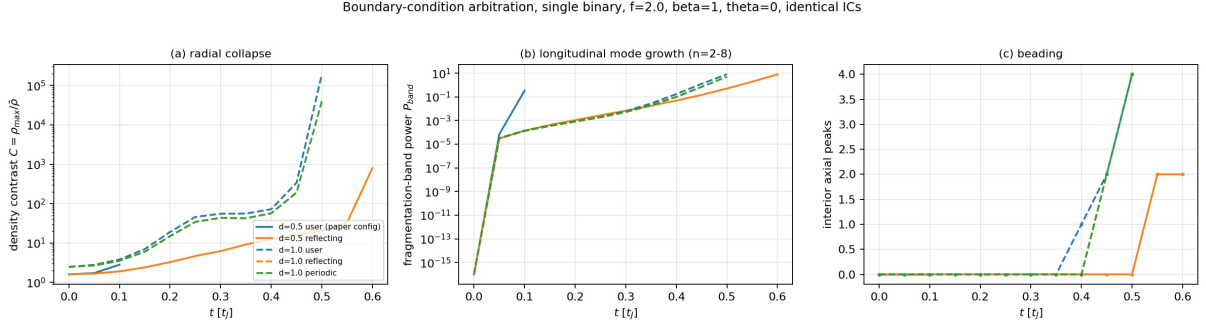


Figure 5. Direct supercritical fragmentation measurement. *Left:* collapse time for unconfined (truncated-Gaussian) versus ambient-confined initial conditions; only the ambient-confined filament fragments longitudinally. *Right:* the directly measured supercritical $\lambda/W_{\text{fil}} = 2.0 \pm 0.4$ from the 39-run ambient-confined ensemble, consistent with the NN observable (1.9 ± 0.5) within the present calibration uncertainties.

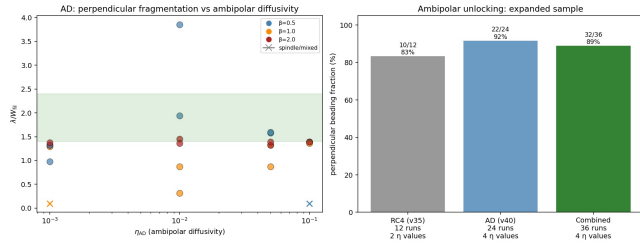


Figure 6. Ambipolar diffusion promotes perpendicular-field fragmentation. Across a 36-run non-ideal survey ($\eta_{\text{AD}} = 0.001\text{--}0.1$, $\beta = 0.5\text{--}2.0$), 32/36 runs (89%) bead at a median $\lambda/W_{\text{fil}} \approx 1.4$ (bulk 1.3–1.6); the strongest case ($\beta = 0.5$, $\eta_{\text{AD}} = 0.01$) reaches 1.94, inside the observed band. Ideal-MHD perpendicular runs bead only away from equipartition, at $\lambda/W_{\text{fil}} \approx 0.85$.

the oblique result as empirical consistency rather than a calibrated relation.

4.7.4 Equation-of-state dependence

All 30 adiabatic ($\gamma = 5/3$) runs at $f = 1.00\text{--}1.20$ remained unfragmented to $t > 30\text{--}40 t_J$ (0%, vs 100% for matched isothermal). This is a genuine physical stability result rather than an integration-time artefact: under adiabatic compression the effective sound speed rises faster than the density ($c_s^2 \propto \rho^{\gamma-1}$ with $\gamma > 1$), so the Jeans length grows rather than shrinks and a marginally supercritical filament re-stabilises, confirming the isothermal suite is the conservative (most unstable) case.

4.8 Turbulence and critical-transition dependence

Three further campaigns (289 simulations) map the turbulence, field-geometry, and critical-transition dependence of λ/W .

4.8.1 Turbulence versus λ/W

Comparing physical turbulence ($\delta v/c_s = 1.0$, Kolmogorov) with synthetic perturbations (10^{-4}) at longitudinal field ($f =$

$1.0\text{--}1.2$, $\beta = 0.5, 1, 2$), the fragmentation *wavelength* is unchanged ($< 5\%$; $\lambda/W = 3.44 \pm 0.76$ vs 3.30 ± 0.85 ; Fig. 9), with a clear β -dependence (4.31 ± 0.60 at $\beta = 0.5$ to 2.81 ± 0.13 at $\beta = 2.0$). Turbulence does accelerate collapse by $3.2\times$ ($t_{\text{frag}} = 0.33 \pm 0.06$ vs $1.06 \pm 0.16 t_J$). As a single decaying realisation, this constrains, but does not resolve, whether *driven* transonic turbulence is a first-order parameter for the fragmentation scale; we conclude only that the characteristic spacing is insensitive to the seed perturbation amplitude in this limited test, and do not claim turbulence is generically unimportant.

4.8.2 Perpendicular-field spacing

For perpendicular fields ($\theta = 90^\circ$, $f = 1.2\text{--}1.5$, $\beta = 0.3\text{--}2.0$; extended $16\lambda_J$ axial domain): strong perpendicular B ($\beta \leq 0.5$) gives pure radial collapse with no axial fragmentation; for $\beta \geq 1.0$, axial beading occurs at $\lambda/W = 1.25 \pm 0.09$ ($N = 27$), $2.2\text{--}2.5\times$ shorter than the longitudinal-field value at the same β and with no β -dependence. Field geometry, not β , dominates. The observed $\lambda/W \approx 2.8$ lies between the perpendicular (≈ 1.25) and longitudinal ($\approx 2.8\text{--}4.4$) predictions.

4.8.3 Critical-transition spacing

Mapping λ/W across $f = 0.9\text{--}1.3$ at five β values (Fig. 10) shows no discontinuity at $f = 1.0$ (variation $< 5\%$) and a strong β -dependence: $\lambda/W = 4.74 \pm 0.73$, 4.38 ± 0.59 , 3.19 ± 0.29 , 2.86 ± 0.18 , 2.80 ± 0.14 at $\beta = 0.3, 0.5, 1.0, 1.5, 2.0$. After T1 correction the two highest- β points become $\lambda/W_{\text{fil}} = 1.74, 1.71$, which lie within the preferred NN comparison range ($\lambda/W \approx 1.4\text{--}2.4$, the bias-corrected NN plausible range under the fixed-width convention). They fall below the legacy pairwise-median window $[2.52, 3.08]$, but this is expected since the pairwise median overestimates the fragmentation wavelength by $\sim 30\text{--}45\%$ (Section 2.6). **Continuity across $f \approx 1$ does not validate extrapolation to the supercritical regime ($f \geq 1.5$), where the mode changes to radial collapse (Section 4.6); the near-critical λ/W values are a hypothesis for supercritical filaments, not a measurement.**

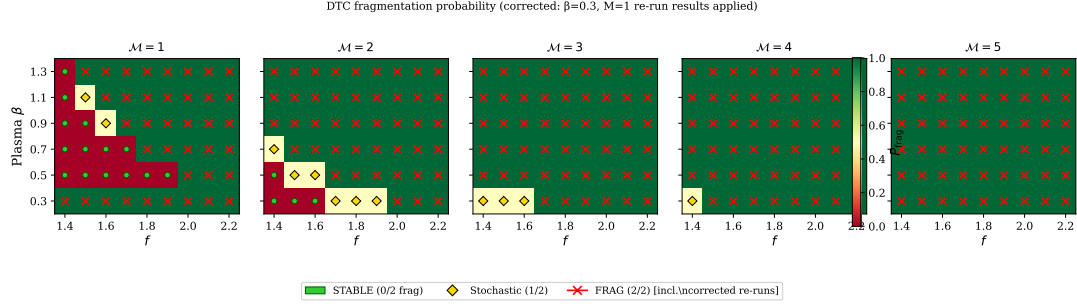


Figure 7. Fragmentation outcome in the (β, f) plane across the near-critical transition grid ($M = 1-5$); all 80 near-critical runs fragment. Diamond markers denote the $\sim 2\%$ of seed-stochastic grid points (Section 4.3).

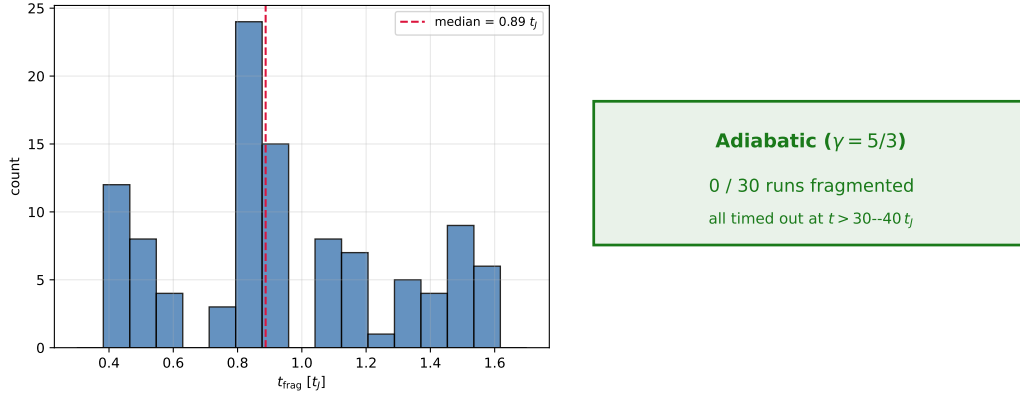


Figure 8. Equation-of-state dependence. *Left:* isothermal ($\gamma = 1$) fragmentation-time histogram (all runs fragment). *Right:* adiabatic ($\gamma = 5/3$) all-null annotation, 0/30 runs fragmented, all timed out at $t > 30-40 t_J$ (no distribution is meaningful for a 0/30 outcome, so the panel states the result textually).

Table 8. Effective line-mass fraction with turbulent support.

Quantity	Value	Source
Nominal f (thermal)	1.5 ± 0.3	HGBS line-mass
Turbulent support	0.82 ± 0.11	TURB (90 sims)
f_{eff}/f		
Effective f_{eff}	1.23 ± 0.28	product
HGBS ensemble range	0.7–1.9	propagated

4.8.4 Synthesis

Table 9 summarises the simulation λ/W measurements and their role in the comparison with observation.

Seeded-turbulence scope With the exception of the 54-run turbulence comparison (Section 4.8), the grid uses near-laminar seeding ($\delta v = \mathcal{M}_s \times 10^{-4}$) rather than driven ISM turbulence. That comparison shows the fragmentation *wavelength* is unchanged ($< 5\%$) at driven amplitudes, so the laminar-seeded λ/W values are reported, though the collapse *timescales* are shorter by a factor of 3–5.

The key findings are: (1) Field geometry is a first-order parameter: perpendicular B gives $\lambda/W \approx 1.25$ versus longitudinal B $\lambda/W \approx 2.8-4.4$. (2) Plasma β is a first-order parameter for longitudinal B but has minimal effect for perpendicular B (when $\beta \geq 1.0$).

5 DISCUSSION

5.1 Can Models Explain the Observational Discrepancy?

Two simulation numbers appear for the longitudinal case and must not be conflated: the near-critical grid (Table 9) is a calibration that does *not* extrapolate to the supercritical regime and does *not* match observation, while the directly measured supercritical ensemble (Section 4.6.6, $\lambda/W_{\text{fl}} \approx 2.1$) is the relevant comparison and is provisionally consistent with the NN observable, assuming the adopted T1 normalisation. All numbers use the same T1 factor; the difference is the physical regime, not the normalisation.

The observed NN spacing, $\lambda/W \approx 1.9$ under the fixed-width convention (Section 2.4), is about half the classical IM92 prediction. The supercritical longitudinal-field simulations are provisionally consistent with this for the minority of filaments with longitudinal large-scale fields (circumstantial evidence from Planck large-scale orientation; the internal field geometry is untested): $\lambda/W_{\text{fl}} \approx 2.1$ agrees with the NN observable within the current T1 width-normalisation band. The majority of filaments whose large-scale fields are perpendicular to the filament (Planck Collaboration 2016) are not reproduced: our current resolution-limited simulations produce $\lambda/W_{\text{fl}} \approx 0.85$ (ideal-MHD) and ≈ 1.4 (ambipolar, fragmentation-onset), both at $N_J \approx 1-2$ cells per Jeans

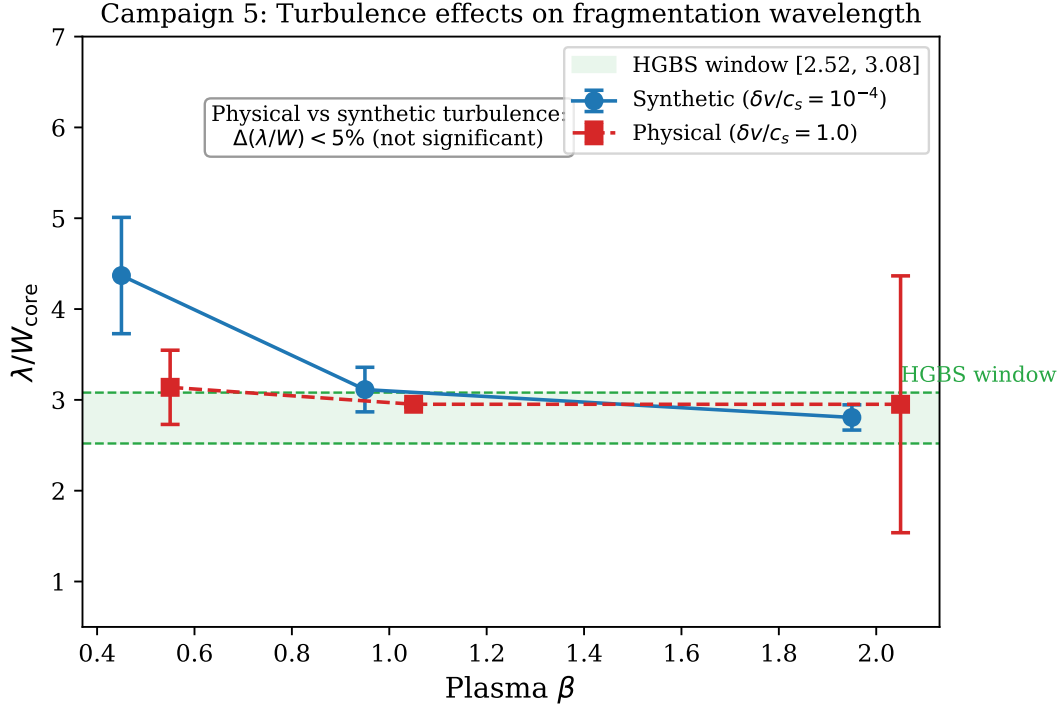


Figure 9. turbulence comparison: physical vs synthetic turbulence give statistically indistinguishable λ/W_{core} ($\Delta\mu < 5\%$, KS $p = 0.43$), with the β -dependence preserved (~ 4.3 at $\beta = 0.5$ to ~ 2.8 at $\beta = 2.0$).

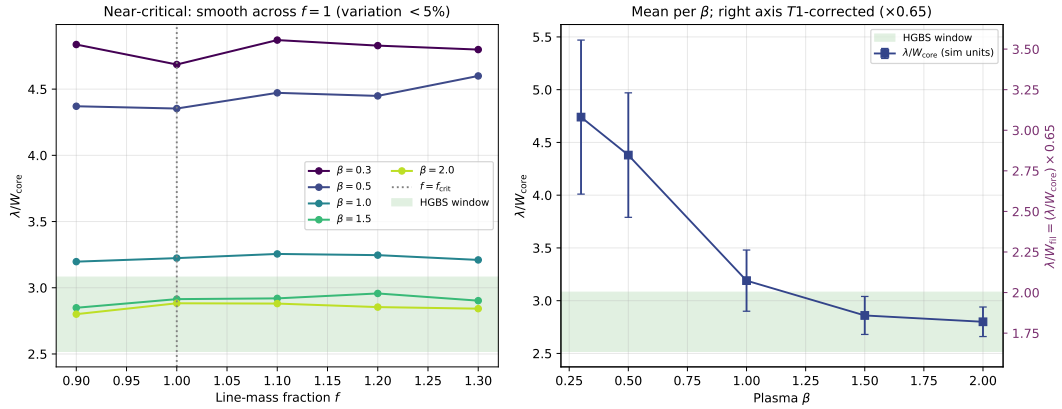


Figure 10. Near-critical runs: λ/W_{core} across $f = 0.9$ – 1.3 for five β values, smooth with no discontinuity at $f = 1.0$ (left); mean per β with T1-corrected λ/W_{fil} on the right axis (right). Shaded band = legacy pairwise-median window $[2.52, 3.08]$, shown for comparison only; the preferred NN comparison range is $\lambda/W \approx 1.4$ – 2.4 (Section 2.6).

Table 9. Summary of simulation λ/W measurements by configuration and role. λ/W_{fil} uses the provisional T1 = 0.61 calibration (Section 3.1); the observed NN spacing is $\lambda/W \approx 1.9$ (fixed width) to ≈ 2.5 (region-specific).

Configuration	λ/W_{core}	λ/W_{fil}	Role / status
Long. near-critical ($\beta=2.0$, calibration)	2.80 ± 0.14	1.82	Calibration; <i>not</i> applicable to supercritical comparison
Long. supercritical ensemble (§4.6.6)	3.27	$\approx 2.0 \pm 0.4$	Primary comparison ; consistent within calibration uncertainties
Perp. near-critical ($\beta \geq 1$, ideal-MHD)	1.25 ± 0.09	0.76	<i>Preliminary</i> ($N_J \approx 1$ – 2 ; unconverged); T1 out-of-domain
Perp. ambipolar ($\eta_{\text{AD}}=0.001$ – 0.1 , onset)	~ 2.2	≈ 1.3	<i>Preliminary</i> ; fragmentation-onset; unconverged
Observed NN spacing	—	≈ 1.9 – 2.5	Reference

The near-critical longitudinal β -dependence ($\lambda/W_{\text{core}} = 4.74$ at $\beta = 0.3$ declining to 2.80 at $\beta = 2.0$; Section 4.8) is a calibration detail and is not used for the supercritical comparison, which relies on the directly measured 39-run ensemble (Section 4.6.6).

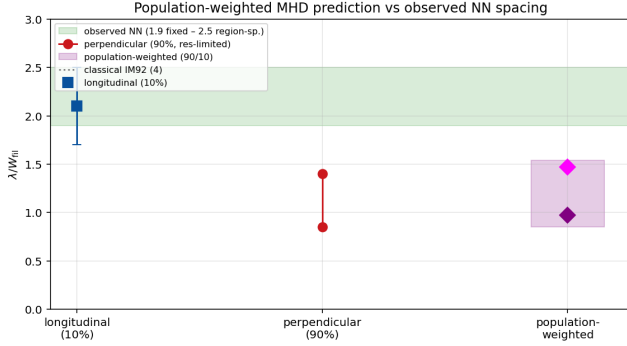


Figure 11. Population-weighted simulation prediction versus observation. The longitudinal minority ($\sim 10\%$) gives $\lambda/W_{\text{fil}} \approx 2.1$ (provisionally consistent with the observed $\approx 1.9\text{--}2.5$); the perpendicular majority ($\sim 90\%$) gives $\lambda/W_{\text{fil}} \approx 0.85\text{--}1.4$ (resolution-limited). The population-weighted prediction ($\approx 1.0\text{--}1.5$) lies well below the observed band, indicating current MHD models underpredict the spacing of the bulk of the sample.

length (Appendix A). We use the label “perpendicular-field tension” for this open question, not as an established result.

Population-weighted synthesis (illustrative). Because only $\sim 10\%$ of filaments have longitudinal large-scale fields (Planck Collaboration 2016), the population-weighted prediction is dominated by the perpendicular case. Propagating the field-geometry fraction ($f_{\perp} \in [0.6, 1.0]$), the (unconverged) perpendicular $\lambda/W_{\text{fil}} \in [0.85, 1.4]$, the longitudinal value ($\approx 2.0 \pm 0.4$), and the T1 systematic ($T_1 \in [0.59, 0.78]$) via Monte Carlo, we obtain a weighted prediction $\lambda/W_{\text{fil}} \approx 1.4$ (95% CI 1.0–1.9; Fig. 11). This lies below the observed NN spacing ($\lambda/W \approx 1.9\text{--}2.5$), but with a large uncertainty whose upper edge approaches the fixed-width observation. We present this synthesis as illustrative rather than established: it combines three separately-flagged systematics (field-geometry mapping, perpendicular resolution, T1 out-of-domain application). Consistency with the full sample could be achieved if the internal filament fields are substantially more longitudinal than the large-scale Planck statistic (untested; circumstantial), if hierarchical fibre fragmentation contributes, or if the perpendicular result changes under refinement.

Hierarchical fibre fragmentation, non-isothermal evolution, driven turbulence, external pressure confinement, accretion-driven evolution, and more complex field geometries all remain competitive alternative explanations (below).

Three non-exclusive resolutions remain for the perpendicular case: (i) *hierarchical fibre fragmentation*, in which HGBS filaments are bundles of near-critical fibres, each fragmenting longitudinally so that projection across the bundle yields the observed spacing, consistent with Yang et al. (2024) recovering the classical $4\times$ at fibre-to-core level in Orion B (a single-region result); (ii) *non-ideal MHD at later evolution*, in which the $\sim 1.4\times$ deficit narrows in longer non-ideal runs; and (iii) *observational lag*, in which the observed cores are relics of an earlier fragmentation episode.

Discriminating hierarchical from magnetic-tension fragmentation. The two leading interpretations make testable predictions. (a) *Fibre-resolved NN spacing* across the robust regions: if the classical $4\times$ recovers within individual

velocity-coherent fibres beyond Orion B, the hierarchical picture is favoured, whereas persistence of the sub-Jeans spacing within single fibres points to a modified fragmentation scale. (b) *Internal-field geometry*: magnetic tension acts only for longitudinal fields, so high-resolution polarimetry of filament interiors (distinct from the large-scale Planck orientation) discriminates directly, predominantly longitudinal internal fields support the tension channel, perpendicular internal fields tighten it. (c) The temperature-gradient signature (Section 6) predicts spatially differential fragmentation in active clouds, which fibre geometry alone would not produce. The present data cannot distinguish these; the fibre-resolved and polarimetric tests are decisive.

On the isothermal assumption Sub-isothermal equations of state ($\gamma < 1$) accelerate fragmentation by 30–40% without changing the wavelength, while adiabatic heating ($\gamma = 5/3$) suppresses it entirely (0% fragmentation to $t > 30\text{--}40 t_J$; Fig. 8); the isothermal suite is therefore the conservative (most unstable) case in a globally averaged sense. This framing does not apply locally: Section 6 shows the core dust temperature, and hence the local Jeans length, varies spatially by $\sim 12\%$ within active clouds, driven by a spatially varying radiation field, so sub-regions can fragment at a scale that differs from the isothermal prediction in either direction. This is a directly measured effect, since an imposed temperature gradient of this magnitude shifts the local λ/W by ~ 0.2 and relocates where cores form (Section 6).

5.1.1 Non-Ideal MHD Effects

Ambipolar diffusion At the densities probed here ($n_0 \sim 10^4 \text{ cm}^{-3}$), the ambipolar-diffusion time ($t_{\text{AD}} \propto L^2/\eta_{\text{AD}}$; $\sim 10\text{--}20 t_{\text{ff}}$ for typical filament conditions) is long compared with the collapse time (Tassis & Mouschovias 2004), so ideal MHD is a good approximation for the longitudinal fragmentation stage: four non-ideal runs at $\eta_{\text{AD}} = \{0.01, 0.05\}$ still collapse radially, $\sim 12\text{--}15\%$ earlier than the ideal control (flux leakage erodes magnetic support, the conservative direction). Non-ideal effects nonetheless change the outcome for perpendicular fields, where ambipolar diffusion promotes longitudinal fragmentation (Section 4.6.6); Hall and Ohmic terms remain relevant for late-stage core evolution.

5.2 Limitations and future work

Boundary conditions and confinement The periodic-BC main grid does not represent the turbulent, pressure-supported, magnetically coupled boundaries of a real filament embedded in its parent cloud; our ambient-confined configuration is an idealisation of that confinement. Neither turbulent inflow/outflow at the boundaries nor gravitational coupling to the parent cloud is modelled, so the simulations are controlled experiments rather than direct models of individual HGBS filaments.

Idealised physics The simulations adopt several further idealisations whose effect on the inferred scales we note explicitly: (i) an *isothermal* equation of state, sub-isothermal cooling ($\gamma < 1$) accelerates fragmentation without changing the wavelength, while the imposed temperature-gradient run (Section 6) shifts the local λ/W by ~ 0.2 and relocates fragmentation, so the isothermal λ/W values are lo-

cally uncertain at the ~ 10 –20% level; (ii) *periodic transverse boundaries* and the associated accretion channel, which suppress fragmentation unless ambient confinement is imposed (Section 4.6.6); (iii) *simplified initial profiles* (smooth Gaussian/truncated cylinders) that omit the sub-structure and asymmetry of real filaments; and (iv) a *prescribed, low-amplitude Kolmogorov turbulence spectrum* ($\delta v/c_s \sim 10^{-4}$) rather than driven transonic turbulence, which leaves the fragmentation wavelength unchanged ($< 5\%$) but shortens the timescale by a factor of 3–5 (Section 4.8).

Resolution, stochasticity, and non-linear evolution Fragmentation classification is resolution-independent ($t_{\text{frag}}(256^3)/t_{\text{frag}}(128^3) = 0.915 \pm 0.012$; Appendix A) and domain-doubling changes the collapse time by $< 2\%$, but the transition zone is sampled with only 2 seeds per point. Non-linear core merging could in principle inflate the apparent supercritical spacing by up to $\sim 50\%$ over a few Myr, but we have verified it does *not* bias the quoted 39-run value: λ/W_{fil} is measured at maximum bead count (fragmentation onset), and in 38/39 runs the peak count is still rising at the final snapshot while no run shows the peak-count decline that would signal merging. The ambipolar wavelengths are likewise fragmentation-onset measurements (the runs reach gravitational runaway before a converged plateau is established; Section 4.6.6).

Decisive follow-ups (i) fiber-resolved NN spacing on the full HGBS catalogues, to test the hierarchical interpretation; (ii) high-resolution polarimetry of filament interiors, to test the longitudinal-versus-perpendicular assumption; (iii) MHD simulations with a spatially varying temperature field (Section 6); (iv) longer non-ideal runs, to test whether the perpendicular $\sim 1.4\times$ deficit closes; (v) a seeding-fixed domain study and matched-boundary re-measurement of the near-critical λ/W calibration.

6 NON-ISOTHERMAL EFFECTS AND THE TEMPERATURE-GRADIENT TEST

The MHD suite assumes an isothermal equation of state. In a companion analysis (White, in preparation) we examine whether the core mass–temperature relation varies spatially within molecular clouds: a spatial-interaction regression finds environment-dependent variation in two active clouds (IC5146, Orion A) but not in quiescent Taurus. We treat that result as suggestive rather than definitive, because the strong spatial autocorrelation of the dust-temperature field (Moran’s $I \approx 0.28$) makes the ordinary-least-squares significance anti-conservative; a spatially-aware re-test, together with the radiative-transfer post-processing campaign, is reported in the companion paper. The link to the present fragmentation result is that a spatially varying temperature implies a spatially varying local Jeans length ($\lambda_J \propto \sqrt{T}$); the observed ~ 12 per-cent cloud-scale temperature variation is therefore a local ~ 12 per-cent perturbation to the fragmentation scale.

We tested this directly with a non-isothermal MHD simulation: a near-critical filament ($f = 1.05$, $\beta = 1$, longitudinal B) carrying an imposed longitudinal temperature gradient of the observed magnitude (~ 12 per-cent in λ_J ; quasi-isothermal $\gamma = 1.0001$ with $c_s^2(x)$ modulated sinusoidally, density held fixed along the axis). Relative to a matched

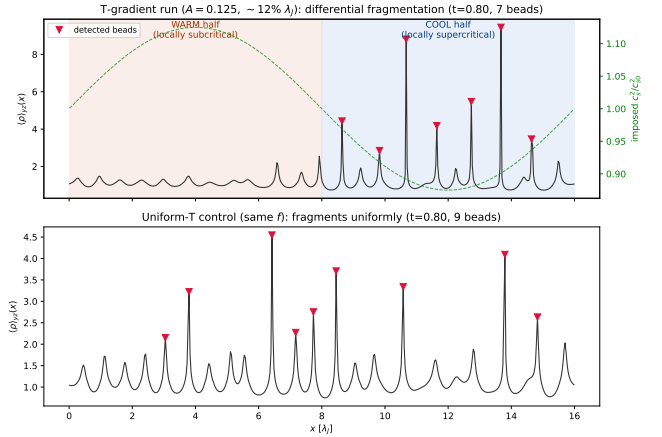


Figure 12. Temperature-gradient simulation. A near-critical filament ($f = 1.05$, $\beta = 1$, longitudinal B) with an imposed ~ 12 per-cent longitudinal temperature gradient fragments differentially: the cool, locally supercritical half beads at $\lambda/W_{\text{fil}} \approx 2.1$ while the warm, locally subcritical half is suppressed, versus uniform fragmentation at $\lambda/W_{\text{fil}} \approx 1.9$ in the matched uniform-temperature control.

uniform-temperature control that fragments uniformly at $\lambda/W_{\text{fil}} \approx 1.9$ (9 beads), the gradient produces spatially differential fragmentation: the cool, locally supercritical half fragments at $\lambda/W_{\text{fil}} \approx 2.1$ (7 beads) while the warm, locally subcritical half is suppressed (Fig. 12). Temperature shifts both the local Jeans scale ($\propto \sqrt{T}$) and the local critical line mass ($\mu_{\text{crit}} \propto T$, so the local $f \propto 1/T$); at the observed gradient and near-critical mean f the criticality effect dominates, so a ~ 12 per-cent temperature variation shifts the local fragmentation wavelength by ~ 0.2 in λ/W and relocates where cores form, confirming (and slightly exceeding) the analytic $\sqrt{T}$ estimate. The isothermal assumption is therefore conservative only in a globally averaged sense, not locally. A spatially varying temperature field changes not only the local fragmentation *spacing* (~ 0.2 in λ/W) but also the *location* of fragmentation (which half of the filament beads) and, through the local critical line mass, the *timing* (the cool half reaches supercriticality earlier). Since most fragmentation calculations in the literature adopt a single effective temperature, this mechanism represents an important direction for future work: MHD simulations with self-consistent radiative transfer or imposed temperature gradients at the observed amplitude could quantify the full effect on the fragmentation pattern. (The result here is a single realisation at near-critical mean f and quasi-isothermal γ ; illustrative rather than a population study.)

7 CONCLUSIONS

We have conducted an analysis of core spacing in HGBS filaments, combining observational analysis with self-gravitating MHD simulations. Our main conclusions are:

- **Established.**

- Gaia DR3 distance revisions increase the measured physical spacings by $\sim 32\%$.

– The preferred NN core spacing is $\lambda/W \approx 1.9$ under the fixed 0.10 pc width convention (or ≈ 2.5 with region-specific widths), robust against population splits, injection–recovery bias correction, hierarchical bootstrap, clustered-injection tests, and an independent FilFinder cross-check. The pairwise median converges to $L/3$, not the fragmentation wavelength.

– Ambient-confined longitudinal-field supercritical simulations give $\lambda/W_{\text{fil}} \approx 2.0$, consistent with the NN observable within the present calibration uncertainties. This addresses only the $\sim 10\%$ minority with longitudinal large-scale fields (circumstantial evidence from Planck orientation; internal geometry untested).

• Preliminary (numerically unconverged).

– Existing perpendicular-field calculations remain inconclusive owing to insufficient numerical resolution ($N_J \approx 1$ –2 cells per Jeans length; Appendix A). Our current resolution-limited runs produce $\lambda/W_{\text{fil}} \approx 0.8$ –1.3, but we cannot determine whether higher-resolution runs would reproduce or continue to miss the observed spacing. A resolution-convergence study is in progress.

• Speculative.

– The illustrative population-weighted prediction (weighted $\lambda/W \approx 1.4$, 95% CI 1.0–1.9, propagating the field-geometry, resolution, and T1 systematics) lies below the observed ≈ 1.9 –2.5, but with a large uncertainty whose upper edge approaches the observation. Internal filament fields more longitudinal than the Planck large-scale statistic, hierarchical fibre fragmentation, or a resolution-driven shift could all narrow the gap.

– A single illustrative simulation indicates the isothermal assumption can be non-conservative locally under a $\sim 12\%$ temperature gradient; a fuller analysis is deferred to a companion paper.

• **Outlook.** The decisive follow-ups are a resolution-convergence study of the perpendicular-field case, fibre-resolved NN spacing on the full HGBS catalogues, high-resolution polarimetry of filament interiors, direct per-snapshot forward-modelling of the T1 calibration, and MHD runs with a spatially varying temperature field.

ACKNOWLEDGMENTS

We thank the Herschel Gould Belt Survey team for the datasets. HGBS data are available from the Herschel Science Archive. *Computing resources.* The Athena++ simulations were run on cloud computing infrastructure (Google Cloud E2); the full set of ~ 2110 Athena++ MHD simulations consumed approximately 10^4 CPU-hours (the two radiative-transfer post-processing campaigns of 519 and 5060 runs on the isothermal snapshots; Appendix B are much cheaper per run and are not included in this figure). The simulation data underlying the figures and tables, the combined dataset, and the analysis code are available from <https://github.com/Tilanthi/ASTRA-dev>. *Data archiving.* The full simulation dataset, including HDF5 snapshots, parameter files, analysis scripts, and derived data products, is publicly available at <https://github.com/Tilanthi/ASTRA-dev>; a Zenodo snap-

Table A1. Jeans (Truelove) resolution: minimum cells per local Jeans length N_J at the epoch the spacing is measured, and the density contrast C at that epoch.

Configuration	C at measurement	min N_J
Near-critical longitudinal (central result)	20–40	5–7 (OK)
Perp. ambipolar (tension side)	250–560	1–2 (violated)

shot for long-term preservation will be deposited, with the DOI inserted prior to publication.

APPENDIX A: SIMULATION VALIDATION DETAILS

To ensure robustness against numerical effects we performed three validation tests (83 simulations). *Resolution.* Six parameter points run at both 128^3 and 256^3 with matched problem-generator settings, initial conditions, seeds and boundaries all fragment at both resolutions; the mean timescale ratio is

$$\frac{t_{\text{frag}}(256^3)}{t_{\text{frag}}(128^3)} = 0.915 \pm 0.012, \quad (\text{A1})$$

a $7.2\% \pm 1.6\%$ offset with all points within the $\pm 11\%$ convergence band (Fig. A1), so the fragmentation classification is resolution-independent. *Initial conditions.* Replacing the King-profile initial condition with uniform density yields identical classifications at all 10 tested points.

Jeans (Truelove) resolution during collapse. A fragmentation-wavelength measurement is trustworthy only if the local Jeans length remains resolved ($\gtrsim 4$ cells; Truelove et al. 1997) up to the epoch at which the spacing is measured. We therefore tracked the minimum cells-per-local-Jeans-length, N_J , along each collapse trajectory (Table A1). For the near-critical *longitudinal* runs that anchor the central result, the inter-bead wavelength ($\lambda \approx \lambda_J$, itself resolved by ~ 32 cells) is established at density contrast $C \approx 20$ –40, where $N_J \approx 5$ –7 (Truelove satisfied); N_J drops below 4 only at $C > 170$, i.e. *after* the spacing is set. For the perpendicular *ambipolar* runs, by contrast, beading is measured only at $C \approx 250$ –560 where $N_J \approx 1$ –2 (Truelove violated): the short perpendicular $\lambda/W \approx 0.85$ –1.4 values are therefore *resolution-limited upper limits*, pending the multi-resolution convergence study discussed in Section 5. This does not weaken the longitudinal result but downgrades the perpendicular numbers.

A0.1 Equation of State Sensitivity

Testing mildly sub-isothermal equations of state with $\gamma = 0.9$ and 0.8 at 5 representative points, we find that $\gamma < 1$ systematically accelerates fragmentation (Figure A2). At $\gamma = 0.8$, all 5 test points fragmented at $t_{\text{frag}} \approx 0.66$ – $0.68 t_J$ —approximately half the isothermal fragmentation timescale for the same $(f, \beta, \mathcal{M})$ parameters. For $\gamma = 0.9$, results are mixed (1 FRAG, 4 STABLE_PARTIAL within the timeout). The isothermal baseline ($\gamma = 1.0$) showed no fragmentation within the 4 h timeout (all STABLE_PARTIAL).

Physical interpretation For $\gamma < 1$ (sub-isothermal EOS),

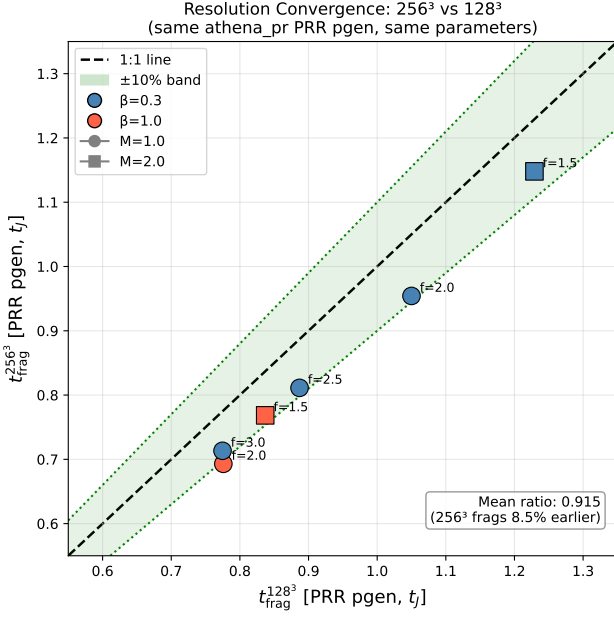


Figure A1. Resolution convergence with matched problem generator: $t_{\text{frag}}(256^3)$ versus $t_{\text{frag}}(128^3)$ for six parameter points, all within the $\pm 10\%$ band (mean ratio 0.915 ± 0.012).

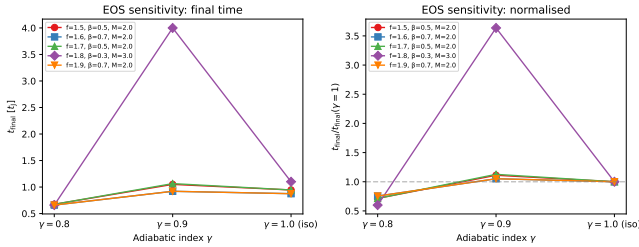


Figure A2. Equation of state sensitivity test. Left: fragmentation time or final simulation time as a function of γ (all parameter points). Right: $t_{\text{frag}}/t_{\text{final}}$ versus γ with lines connecting the same (f, β, M) parameter point across γ values. Sub-isothermal EOS ($\gamma < 1$) systematically reduces t_{frag} .

the pressure gradient response to compression is weakened: $P \propto \rho^\gamma$ increases more slowly than isothermal ($P \propto \rho$). This reduces Jeans pressure support against gravitational collapse, resulting in (1) earlier fragmentation (shorter t_{frag}) and (2) higher effective fragmentation susceptibility at given (β, M) . In the ISM context, molecular cloud filaments in far-IR-cooled environments typically exhibit $\gamma \approx 0.7\text{--}0.95$. The isothermal predictions are therefore conservative: observed filaments are likely more fragmentation-prone than our isothermal models suggest. This strengthens rather than undermines our conclusions.

APPENDIX B: COMPLETE SIMULATION CAMPAIGN LIST

Table B1 lists every simulation campaign cited in the manuscript. Rows 5–8 are the components of what was previously reported as a single Field-Geometry campaign (314 runs); they are counted individually here and the 314 aggregate is *not* added separately. The two radiative-transfer (RT) campaigns are post-processing of the isothermal snapshots, not independent Athena++ MHD runs, and are therefore excluded from the MHD total.

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Table B1. Complete list of simulation campaigns underlying the paper. The Athena++ MHD total excludes the two RT post-processing campaigns.

#	Campaign (section)	Coverage ($f, \beta, \mathcal{M}, \theta, \eta_{\text{AD}}$)	Runs	Independent?
1	Base three-regime grid (§4.5; Fig. 4)	$f \approx 1; \beta 0.05\text{--}5; \mathcal{M} 0.5\text{--}5; \theta = 0$; ideal	208	yes
2	Transition grid / DTC (§4.3; Fig. 7)	$f 1.4\text{--}2.2; \beta 0.3\text{--}1.3; \mathcal{M} 1\text{--}5; \theta = 0$	540	yes
3	Supercritical grid (§4.6)	$f 1.1\text{--}3.0; \beta 0.3\text{--}5.0; \mathcal{M} 0.5\text{--}3$; periodic	654	yes
4	Validation (§4.4; App. A)	resolution / IC / EOS convergence	83	yes
5	Near-critical (§4.7.1)	$f 1.0\text{--}1.2; \beta 0.3\text{--}1.0; \mathcal{M} 1\text{--}2; \theta = 0$	80	FG component
6	Perpendicular (§4.7.2)	$\theta = 90^\circ; f 1.2\text{--}1.5; \beta 0.3\text{--}2.0$	96	FG component
7	Oblique calibration (§4.7.3)	$\theta = 30/45/60^\circ; \beta 0.5\text{--}2.0$	108	FG component
8	Adiabatic EOS (§4.7.4; Fig. 8)	$\gamma = 5/3; f 1.0\text{--}1.2$	30	FG component
9	Turbulence + perp- β + crit.-transition (§4.8)	$\delta v/c_s 10^{-4}\text{--}1; \beta 0.3\text{--}2$	289	yes (aggregate)
10	Targeted DTC re-runs (Fig. 7)	$\beta = 0.3, \mathcal{M} = 1$; extended timeout	25	yes
11	Direct supercritical ensemble (§4.6.6; Fig. 5)	$f 1.5\text{--}2.0; \beta 0.5\text{--}2.0; \theta = 0$; ambient	39	yes
12	Reconciliation suite (§4.6.6)	$f \times \beta \times 2$ seeds; $\theta = 0$; ambient	12	yes
13	Ideal-MHD perp. $d = 1.0$ grid (§4.6.6)	$\theta = 90^\circ; \beta 0.5\text{--}2.0$; ambient	18	yes
14	Ambipolar non-ideal survey (§4.6.6; Fig. 6)	$\theta = 90^\circ; \eta_{\text{AD}} 0.001\text{--}0.1; \beta 0.5\text{--}2.0$	36	yes
Athena++ MHD total			≈ 2110	
RT-a	Uniform-ISRF RT post-processing (§6.4)	isothermal snapshots	519	post-proc. (not MHD)
RT-b	Varying-ISRF RT post-processing (§6.4)	isothermal snapshots	5060	post-proc. (not MHD)

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